

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\161-C1-C2 TOP.pvdb

Date of Analysis : Sep 9, 2026 9:39am

PV Elite 28, January 2026



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FileName : 161-C1-C2 _(TOP)

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Vessel Design Summary:

Step: 31

9:39am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 188.951 mm ID

Vessel Design Length, Tangent to Tangent 6786.95 mm

Specified Datum Line Distance 50.80 mm

Shell Side Design Temperature 260 °C

Channel Side Design Temperature 270 °C

Shell Side Design Pressure 2400.000 KPa.

Channel Side Design Pressure 2400.000 KPa.

Shop Shell Side Test Pressure 13260.004 KPa.

Field Shell Side Test Pressure 3120.000 KPa.

Shop Channel Side Test Pressure 10387.897 KPa.

Field Channel Side Test Pressure 3120.000 KPa.

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-106 B	...	...	K03006	
Head	SA-516 70	...	...	K02700	
Flange	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS dish end	2401.843	103.42	14749.000	3.0000	No	No
LHS Ch Shell	2401.843	103.42	11594.000	3.0000	N/A	No
LHS Ch Flange	2401.911	103.42	10275.000	3.0000	N/A	No
Main Shell	2401.843	103.42	11594.000	3.0000	N/A	No
RHS Ch Flg	2401.911	103.42	4588.000	3.0000	N/A	No
RHS Ch Shell	2401.843	103.42	11594.000	3.0000	N/A	No
RHS Dish End	2401.843	103.42	8345.000	3.0000	No	No

Liquid Level: 195.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
---------	-----------	---------	---------	----------	----------	----------	------	------



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Vessel Design Summary:

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Type	mm	Length mm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Ellipse	0.0	50.8	15.1	13.2	4.6	3.6	1.00	1.00
Cylinder	300.0	300.0	15.1	13.2	5.0	3.9	1.00	1.00
Body Flg	385.2	82.0	68.0	56.0	48.4	48.4	1.00	1.00
Cylinder	6240.2	5785.0	15.1	13.2	5.0	4.9	1.00	1.00
Body Flg	6385.4	72.0	64.0	56.0	49.4	49.4	1.00	1.00
Cylinder	6685.4	300.0	15.1	13.2	5.0	3.9	1.00	1.00
Ellipse	6736.2	50.8	10.3	9.0	4.7	3.6	1.00	1.00

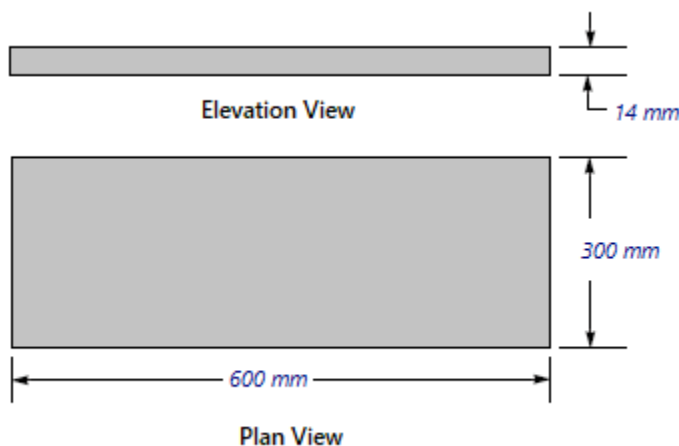
Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.656	Shell	Passed
N3-Shell inlet	WRC-297	0.656	Shell	Passed
N4-Shell outle	WRC-297	0.656	Shell	Passed
N2-Tube Outlet	WRC-297	0.656	Shell	Passed

Foundation/Support Summary**Saddle Parameters:**

Saddle Width	275.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	535.000 mm
Wear Pad Width	330.000 mm
Wear Pad Thickness	14.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1200.000 mm

Baseplate Length	600.000 mm
Baseplate Thickness	14.000 mm
Baseplate Width	300.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	14.000 mm
Web Thickness	14.000 mm
Height of Center Web	440.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



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Vessel Design Summary:

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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	1609.19	Kgf
Maximum Transverse Saddle Shear Load	142.31	Kgf
Maximum Longitudinal Saddle Shear Load	99.62	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1145.39	Kgf
Maximum Transverse Saddle Shear Load	40.87	Kgf
Maximum Longitudinal Saddle Shear Load	8.38	Kgf

Weights:

Fabricated - Bare w/o removable internals	1481.8	kg.
Shop Test - Fabricated + water (full)	1639.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	1481.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1481.8	kg.
Empty - Fab. + intls. + details + wghts.	1727.0	kg.
Operating - Empty + operating liquid (no ca)	1884.2	kg.
Field Test - Empty weight + test medium (full)	1781.2	kg.

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Nozzle Summary:

Step: 29

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	11594	...	...	OK	5.01	OK	No Calc[*]
N3-Shell inlet	11594	...	...	OK	5.01	OK	No Calc[*]
N4-Shell outlet	11594	...	...	OK	5.01	OK	No Calc[*]
N2-Tube Outlet	11594	...	...	OK	5.01	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 11594 Nozzle : N3-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 11594 Nozzle : N1-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1-Tube Inlet	200.800	90.000	146.000
50	N3-Shell inlet	1105.975	90.000	146.000
50	N4-Shell outlet	5755.975	270.000	146.000
80	N2-Tube Outlet	6586.150	270.000	146.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 28 9:39am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	89 mm	Actual	WNF	89.0	11.000	129.00	14.00	224	600
N3-Shell inlet	89 mm	Actual	WNF	89.0	11.000	129.00	12.00	224	600
N4-Shell outlet	89 mm	Actual	WNF	89.0	11.000	129.00	12.00	224	600
N2-Tube Outlet	89 mm	Actual	WNF	89.0	11.000	129.00	12.00	224	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-106 B	10.000	12.000	10.000	14.000	...
N3-Shell in	SA-106 B	11.000	10.000	10.000	12.000	...
N4-Shell ou	SA-106 B	11.000	10.000	10.000	12.000	...
N2-Tube Out	SA-106 B	11.000	10.000	10.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	150.000	90.0	200.00	0.00	LHS Ch Shell
N3-Shell inlet	1055.175	90.0	200.00	0.00	Main Shell
N4-Shell outlet	5705.175	270.0	200.00	0.00	Main Shell
N2-Tube Outlet	6535.350	270.0	200.00	0.00	RHS Ch Shell

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 15.1mm THK X 189.0mm ID X 50.8mm	SA-516 70
2	CYLINDER: 15.1mm THK X 189.0mm ID X 300.0mm	SA-106 B
1	WELD NECK FLANGE: 68.0mm THK X 360.0mm OD	SA-105
1	CYLINDER: 15.1mm THK X 189.0mm ID X 5785.0mm	SA-106 B
1	WELD NECK FLANGE: 64.0mm THK X 360.0mm OD	SA-105
1	ELLIPTICAL HEAD: 2.0 X 1, 10.3mm THK X 198.4mm ID X 50.8mm	SA-516 70
2	INSULATION: 50mm X 100mm THK	Hot Insulation
2	INSULATION: 300mm X 100mm THK	Hot Insulation
2	INSULATION: 83mm X 100mm THK	Hot Insulation
10	BAFFLES: 204.0mm DIA X 6.0mm THK	SA-516 70
4	TIE RODS: 10mm DIA X 5000mm	SA-193 B7M
2	TUBESHEET: 360.0mm X 70.0mm THK	SA-266 2
81	TUBES: 6000.0mm X 12.7mm DIA X 2.1mm THK	SA-179
2	SADDLE: 275.0mm X 120 DEG	SA-516 70
1	INSULATION: 5780mm X 100mm THK	Hot Insulation
1	GASKET: 247.0mm OD X 221.0mm ID	...
24	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
24	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
1	GASKET: 260.7mm OD X 234.7mm ID	...
1	NAMEPLATE	...



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MDMT Summary:

Step: 30 9:39am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main Shell [8]	B	-20	-104	-29	0.201	13.179	1.00	No	
LH floating s[24]	B	-20	-48	-29	0.417	13.179	1.00	No	
RH fixed sadd[24]	B	-20	-48	-29	0.417	13.179	1.00	No	
N3-Shell inlet[1]	B	-5	-104	-29	0.112	11.000	1.00	No	
Nozzle Flg [5]	A	-18	-104						
N4-Shell outle[1]	B	-5	-104	-29	0.112	11.000	1.00	No	
Nozzle Flg [5]	A	-18	-104						
Tubesheet: SS[13]	B	-11	-104	-29	0.234	17.500	1.00	No	
Warmest MDMT:			-5	-48					
LHS Ch Flange[11]	A	0	-104		0.338	13.179	1.00	No	
RHS Ch Flg [11]	A	-8	-104	-29	0.338	7.156	1.00	No	
LHS dish end [10]	B	-20	-104	-29	0.307	13.179	1.00	No	
LHS dish end [7]	B	-16	-104	-29	0.274	15.062	1.00	No	
LHS Ch Shell [8]	B	-20	-104	-29	0.185	13.179	1.00	No	
RHS Ch Shell [8]	B	-20	-104	-29	0.185	13.179	1.00	No	
RHS Dish End [10]	B	-29	-48	-29	0.545	9.023	1.00	No	
RHS Dish End [7]	B	-28	-48	-29	0.474	10.312	1.00	No	
N1-Tube Inlet [1]	B	-5	-104	-29	0.112	11.000	1.00	No	
Nozzle Flg [5]	A	-18	-104						
N2-Tube Outlet[1]	B	-5	-104	-29	0.112	11.000	1.00	No	
Nozzle Flg [5]	A	-18	-104						
Tubesheet: CS[14]	B	-11	-104	-29	0.234	17.500	1.00	No	
Bolting [21]		-48							
Bolting [21]		-48							
Warmest MDMT:			0	-48					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-48.0	-12.0	Pass
Channel/Tube	-48.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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MDMT Summary:

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UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:39am Sep 9,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2400	KPa.
Channel Side Design Pressure	2400	KPa.
Shell Side Design Temperature	260.0	°C
Channel Side Design Temperature	270.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:39am Sep 9,2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	LHS dish end	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	188.95	mm
Element Thickness	13.179	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	15.062	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2400	KPa.
Design Temperature Internal Pressure	270	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	270	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	137.2	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	185	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	10	
Detail Type	Insulation	



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Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.8 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	300 mm
Inside Diameter	188.95 mm
Element Thickness	13.179 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	15.062 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	216 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	194.8 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	185 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	300 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	89 mm
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	23.982 Kg
Grade of Attached Flange	GR 1.1



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Nozzle Matl

SA-106 B

Element#	3/7
Element From Node	30
Element To Node	50
Element Type	Flange
Description	LHS Ch Flange
Distance "FROM" to "TO"	82 mm
Flange Inside Diameter	188.95 mm
Element Thickness	56 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-105
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	133.2 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	200 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K03504
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	195 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	83 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	4/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	5785 mm
Inside Diameter	188.95 mm
Element Thickness	13.179 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	15.062 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	260 °C
Design External Pressure	103.42 KPa.



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Input Echo:

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Design Temperature External Pressure	260	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	216	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	196.4	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	LH floating sad	
Dist. from "FROM" Node / Offset dist	1200	mm
Width of Saddle	275	mm
Height of Saddle at Bottom	535	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	330	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	RH fixed saddle	
Dist. from "FROM" Node / Offset dist	4650	mm
Width of Saddle	275	mm
Height of Saddle at Bottom	535	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	330	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	185	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	5780	mm



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Input Echo:

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Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3-Shell inlet	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	89	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.851	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4-Shell outlet	
Dist. from "FROM" Node / Offset dist	5250	mm
Nozzle Diameter	89	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.851	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	5/7	
Element From Node	60	
Element To Node	80	
Element Type	Flange	
Description	RHS Ch Flg	
Distance "FROM" to "TO"	72	mm
Flange Inside Diameter	202.72	mm
Element Thickness	56	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	64	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2400	KPa.
Design Temperature Internal Pressure	260	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	260	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-105	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	134.6	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	202	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K03504	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm



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Input Echo:

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Height/Length of Liquid	195	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	83	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	6/7	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	RHS Ch Shell	
Distance "FROM" to "TO"	300	mm
Inside Diameter	188.95	mm
Element Thickness	13.179	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	15.062	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2400	KPa.
Design Temperature Internal Pressure	270	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	270	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	216	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	194.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	185	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	300	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	89	mm
Nozzle Schedule	160	



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Input Echo:

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Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	23.851 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	7/7
Element From Node	90
Element To Node	100
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	198.45 mm
Element Thickness	9.0233 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	10.312 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	137.2 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	211.2 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	185 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	90
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.8 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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XY Coordinate Calculations:

Step: 2 9:39am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS dish end		50.8	...	50.8	...
LHS Ch Shell		350.8	...	300	...
LHS Ch Flange		435.975	...	82	...
Main Shell		6290.98	...	5785	...
RHS Ch Flg		6436.15	...	72	...
RHS Ch Shell		6736.15	...	300	...
RHS Dish End		6786.95	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

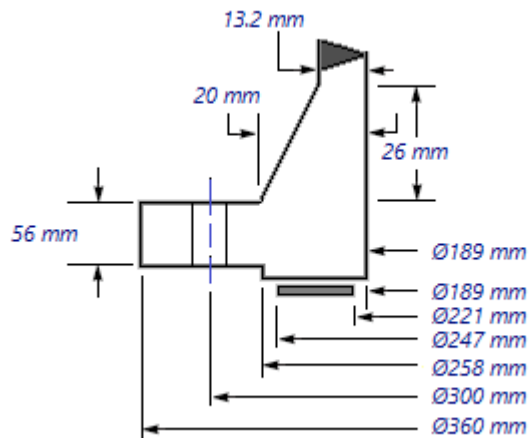
Flng: 5 9:39am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	2401.91	KPa.
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	188.951	mm
Flange Outside Diameter	A	360.000	mm
Flange Thickness	t	56.0000	mm
Thickness of Hub at Small End	go	13.1794	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	26.0000	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	133.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	300.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	UNC Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	257.951	mm
Flange Face Inside Diameter	Fid	188.951	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	246.951	mm
Gasket Inside Diameter	Gi	220.951	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2402 \cdot (189/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 2402) + Ca$$

$$= 5.0088 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 10.2) / (97.5 + 0.6 \cdot 10.2)$$

$$= 11595.303 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 13.2) / (94.5 + 0.6 \cdot 13.2)$$

$$= 15188.544 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	53.000 mm
Corroded Flange ID, Bcor = B+2*ci	194.951 mm
Corroded Large Hub, g1Cor = g1-ci	17.000 mm
Corroded Small Hub, g0Cor = go-ci	10.179 mm
Code R Dimension, R = ((C-Bcor)/2) - g1cor	35.525 mm

Gasket Contact Width, N = (Go - Gi)/2	13.000 mm
Basic Gasket Width, bo = N/2	6.500 mm
Effective Gasket Width, b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter, G = Go - 2 * b	234.102 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 234^2 \cdot 2402$$

$$= 10542.734 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 234 \cdot 3 \cdot 2402$$

$$= 6943.897 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 195^2 \cdot 2402$$

$$= 7311.295 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 10543 - 7311$$

$$= 3231.438 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(10543 + 6944 + 0, 0)$$

$$= 17486.631 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 234 + 0 \cdot 0$$

$$= 33220.336 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((17487 + 0 + 4 \cdot 0 / 234) / 172, 33220 / 172)$$

$$= 18.941 \text{ cm}^2$$



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Flg Calc [Int P]: New Flange

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 53/(3 + 0.5)$$

$$= 141.657 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 300 \cdot \sin(\pi/12)$$

$$= 77.646 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{77.6/(2 \cdot 25.4 + 53)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	18.941	42.658	
Radial Distance between Hub and Bolts:	34.925	35.525	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	77.646	141.657

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (247 + 221))$$

$$= 7.239 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 17487 - 10543$$

$$= 6943.90 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (300 - 234) / 2$$

$$= 32.9491 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((300 - 195) / 2 + 32.9)$$

$$= 42.7368 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (300 - 195 - 17) / 2$$

$$= 44.0245 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	7311.	44.0245	1.0000	322.
Face Pressure, Mt	3231.	42.7368	1.0000	138.
Gasket Load, Mg	6944.	32.9491	1.0000	229.
Gasket Seating, Matm	74816.	32.9491	1.0000	2465.



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Flg Calc [Int P]: New Flange

Flng: 5 9:39am Sep 9, 2026

Total Moment for Operation, Mo 689. Kg-m.
Total Moment for Gasket seating, Mg 2465. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 44.548 mm
Hub Ratio, h/h0 = HL / H0 0.584
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.670

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.823
Factor V per 4.16.5 0.259
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 360/195
= 1.847

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.85² • log10(1.85))) / (1.85² - 1)) / (1.85 - 1)
= 3.335

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.85²(1 + 8.55246 • log10(1.85)) - 1) / ((1.0472 + 1.9448 • 1.85²) • (1.85 - 1))
= 1.566

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.85²(1 + 8.55246 • log10(1.85)) - 1) / (1.36136(1.85² - 1) • (1.85 - 1))
= 3.664

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.85² + 1) / (1.85² - 1)
= 1.830

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 3.66 • 10.2² • 44.5 / 0.26
= 65359.688 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.82/44.5
= 0.018 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (53 • 0.018 + 1) / 1.57 + 53³/65360
= 3.542

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1•689/195)/(3.54•17²)
= 33.85 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1•2465/195)/(3.54•17²)
= 121.15 N./mm²



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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.31 \cdot 689 / 195) / (3.54 \cdot 53^2)$$

$$= 8.03 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.31 \cdot 2465 / 195) / (3.54 \cdot 53^2)$$

$$= 28.73 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (3.33 \cdot 689 / (53^2 \cdot 195)) - 1.83 \cdot 8.03$$

$$= 26.44 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (3.33 \cdot 2465 / (53^2 \cdot 195)) - 1.83 \cdot 28.7$$

$$= 94.63 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (33.8 + \max(8.03, 26.4)) / 2$$

$$= 30.15 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (121 + \max(28.7, 94.6)) / 2$$

$$= 107.89 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 17487 / 42.7$$

$$= 40.20 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (33220 / 42.7)$$

$$= 76.37 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	33.85	199.80	121.15	207.00
Radial Flange	8.03	133.20	28.73	138.00
Tangential Flange	26.44	133.20	94.63	138.00
Maximum Average	30.15	133.20	107.89	138.00
Bolting	40.20	172.00	76.37	172.00

Minimum Required Flange Thickness 48.387 mm

Estimated M.A.W.P. (Operating) 10276 KPa.

Estimated Finished Weight of Flange at given Thk. 34.2 kg.

Estimated Unfinished Weight of Forging at given Thk 46.9 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub Sh	384.99	457.15	0.842	384.99	499.99	0.770



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Tangential Flange	St	300.73	342.86	0.877	300.73	374.99	0.802
Tangential Flange	Sto	142.05	228.57	0.621	142.05	250.00	0.568
Combined	Sh/f + St	685.72	685.72	1.000	685.72	749.99	0.914
Combined	Sr + St	392.04	685.72	0.572	392.04	749.99	0.523
Combined	Sh + Sto	527.04	685.72	0.769	527.04	749.99	0.703
Radial Flange	Sr	91.31	457.15	0.200	91.31	499.99	0.183
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.2/1.86)

= 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.79/2.03) if $h/h_0 < 0.35$, else max (0.25; 0.29/0.78)

= 0.3690

In this case, the hub ratio h/h_0 , 0.5836 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:



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Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	546.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	3.5548 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	200.0 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.167 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (247^2 - 221^2)$$

$$= 95.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207 \cdot 95.5 / (12 \cdot 3.55)$$

$$= 463.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(464; 508)$$

$$= 463.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(464; 290)$$

$$= 463.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(464; 547)$$

$$= 463.6 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 464 \geq 140 (95.5 / (12 \cdot 3.55))$$

$$\therefore 463.6 \geq 313.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 464 \geq (97 \cdot 95.5 + \pi/4 \cdot 2402 \cdot 221^2) / (0.7 \cdot 3.55 \cdot 12)$$

$$\therefore 463.6 \geq 341.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 464 \leq 276 (95.5 / (12 \cdot 3.55))$$

$$\therefore 463.6 \leq 618.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$\therefore 464 \leq 547 (1/0.17)$$

$$\therefore 463.6 \leq 3281.2 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$



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Flg Calc [Int P]: New Flange

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$$= 464 \cdot 0.2 \cdot 3.55 \cdot 25.4$$

$$= 85.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2465 / 1 \cdot 9807 \cdot 0.26 / (2.6 \cdot 202713 \cdot 10.2^2 \cdot 44.5 \cdot 0.3)$$

$$= 0.446 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 689 / 1 \cdot 9807 \cdot 0.26 / (2.6 \cdot 187243 \cdot 10.2^2 \cdot 44.5 \cdot 0.3)$$

$$= 0.135 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2465 / 1 \cdot 9807 \cdot 0.26 / (3.54 \cdot 202713 \cdot 10.2^2 \cdot 44.5 \cdot 0.3)$$

$$= 0.328 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 689 / 1 \cdot 9807 \cdot 0.26 / (3.54 \cdot 187243 \cdot 10.2^2 \cdot 44.5 \cdot 0.3)$$

$$= 0.099 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.34, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	30.13	207.00	105.48	207.00
Radial Flange	8.69	138.00	30.42	138.00
Tangential Flange	21.74	138.00	76.14	138.00
Maximum Average	25.93	138.00	90.81	138.00
Bolting	40.20	172.00	76.37	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 41.097 mm

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Flg Calc [Int P]: New Flange

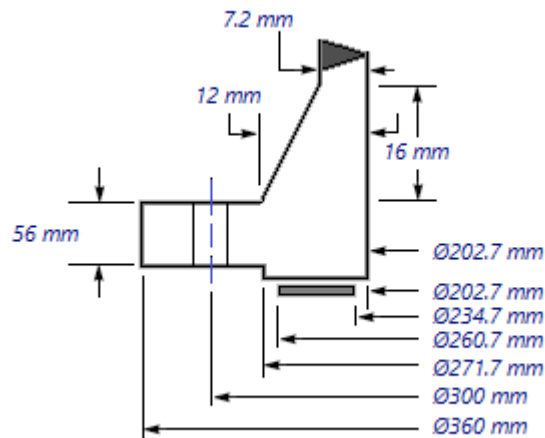
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Flange Input Data Values

Description: New Flange :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	2401.91	KPa.
Design Temperature		260	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	202.717	mm
Flange Outside Diameter	A	360.000	mm
Flange Thickness	t	56.0000	mm
Thickness of Hub at Small End	go	7.1564	mm
Thickness of Hub at Large End	gl	12.0000	mm
Length of Hub	h	16.0000	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	134.60	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	300.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	UNC Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	271.717	mm
Flange Face Inside Diameter	Fid	202.717	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	260.717	mm
Gasket Inside Diameter	Gi	234.717	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2402 \cdot (203/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 2402) + Ca$$

$$= 5.1507 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 4.16) / (104 + 0.6 \cdot 4.16)$$

$$= 4589.732 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 7.16) / (101 + 0.6 \cdot 7.16)$$

$$= 7992.219 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	53.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	208.717 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	9.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	4.156 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	36.641 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	247.868 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 248^2 \cdot 2402$$

$$= 11819.086 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 248 \cdot 3 \cdot 2402$$

$$= 7352.222 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 209^2 \cdot 2402$$

$$= 8380.291 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 11819 - 8380$$

$$= 3438.796 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(11819 + 7352 + 0, 0)$$

$$= 19171.309 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 248 + 0 \cdot 0 \cdot 0$$

$$= 35173.809 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((19171 + 0 + 4 \cdot 0/248)/172, 35174/172)$$

$$= 20.055 \text{ cm}^2$$



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Flg Calc [Int P]: New Flange

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 53/(3 + 0.5)$$

$$= 141.657 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 300 \cdot \sin(\pi / 12)$$

$$= 77.646 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{77.6/(2 \cdot 25.4 + 53)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	20.055	42.658	
Radial Distance between Hub and Bolts:	34.925	36.642	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	77.646	141.657

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (261 + 235))$$

$$= 6.837 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 19171 - 11819$$

$$= 7352.22 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (300 - 248) / 2$$

$$= 26.0661 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((300 - 209) / 2 + 26.1)$$

$$= 35.8538 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (300 - 209 - 9) / 2$$

$$= 41.1415 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8380.	41.1415	1.0000	345.
Face Pressure, Mt	3439.	35.8538	1.0000	123.
Gasket Load, Mg	7352.	26.0661	1.0000	192.
Gasket Seating, Matm	74816.	26.0661	1.0000	1950.



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Flg Calc [Int P]: New Flange

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Total Moment for Operation, Mo 660. Kg-m.
Total Moment for Gasket seating, Mg 1950. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 29.454 mm
Hub Ratio, h/h0 = HL / H0 0.543
Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.165

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.809
Factor V per 4.16.5 0.200
Factor f per 4.16.5 1.280

Stress factor from Table 4.16.4 [K]:

= A / B
= 360/209
= 1.725

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.72² • log10(1.72)) / (1.72² - 1)) / (1.72 - 1)
= 3.735

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.72²(1 + 8.55246 • log10(1.72)) - 1) / ((1.0472 + 1.9448 • 1.72²) • (1.72 - 1))
= 1.615

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.72²(1 + 8.55246 • log10(1.72)) - 1) / (1.36136(1.72² - 1) • (1.72 - 1))
= 4.104

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.72² + 1) / (1.72² - 1)
= 2.013

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 4.1 • 4.16² • 29.5 / 0.2
= 10445.670 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.81/29.5
= 0.027 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (53 • 0.027 + 1) / 1.62 + 53³/10446
= 15.773

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.28•660/209)/(15.8•9²)
= 31.07 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1.28•1950/209)/(15.8•9²)
= 91.83 N./mm²



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Flg Calc [Int P]: New Flange

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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.94 \cdot 660 / 209) / (15.8 \cdot 53^2)$$

$$= 2.06 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.94 \cdot 1950 / 209) / (15.8 \cdot 53^2)$$

$$= 6.08 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (3.73 \cdot 660 / (53^2 \cdot 209)) - 2.01 \cdot 2.06$$

$$= 37.07 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (3.73 \cdot 1950 / (53^2 \cdot 209)) - 2.01 \cdot 6.08$$

$$= 109.59 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (31.1 + \max(2.06, 37.1)) / 2$$

$$= 34.07 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (91.8 + \max(6.08, 110)) / 2$$

$$= 100.71 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 19171 / 42.7$$

$$= 44.07 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (35174 / 42.7)$$

$$= 80.86 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	31.07	201.90	91.83	207.00
Radial Flange	2.06	134.60	6.08	138.00
Tangential Flange	37.07	134.60	109.59	138.00
Maximum Average	34.07	134.60	100.71	138.00
Bolting	44.07	172.00	80.86	172.00

Minimum Required Flange Thickness 49.403 mm

Estimated M.A.W.P. (Operating) 4590 KPa.

Estimated Finished Weight of Flange at given Thk. 31.0 kg.

Estimated Unfinished Weight of Forging at given Thk 38.8 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub Sh	293.32	461.72	0.635	293.32	499.99	0.587



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Flg Calc [Int P]: New Flange

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Tangential Flange	St	346.29	346.29	1.000	346.29	374.99	0.923
Tangential Flange	Sto	132.11	230.86	0.572	132.11	250.00	0.528
Combined	Sh/f + St	572.90	692.57	0.827	572.90	749.99	0.764
Combined	Sr + St	365.51	692.57	0.528	365.51	749.99	0.487
Combined	Sh + Sto	425.44	692.57	0.614	425.44	749.99	0.567
Radial Flange	Sr	19.22	461.72	0.042	19.22	499.99	0.038
Maximum Stress Ratio				1.000			0.923

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 2.2/1.7)

= 1.2944

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 1.08/1.75) if h/ho < 0.35, else max (0.25; 0.45/0.77)

= 0.5830

In this case, the hub ratio h/ho, 0.5432 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:



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Flg Calc [Int P]: New Flange

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Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	543.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	3.5548 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	202.0 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.101 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (261^2 - 235^2)$$

$$= 101.2 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 101 / (12 \cdot 3.55)$$

$$= 490.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(491; 508)$$

$$= 490.9 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(491; 290)$$

$$= 490.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(491; 543)$$

$$= 490.9 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 491 \geq 140 (101 / (12 \cdot 3.55))$$

$$:: 490.9 \geq 332.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 491 \geq (97 \cdot 101 + \pi/4 \cdot 2402 \cdot 235^2) / (0.7 \cdot 3.55 \cdot 12)$$

$$:: 490.9 \geq 363.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 491 \leq 276 (101 / (12 \cdot 3.55))$$

$$:: 490.9 \leq 654.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$:: 491 \leq 543 (1/0.1)$$

$$:: 490.9 \leq 5380.9 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$



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Flg Calc [Int P]: New Flange

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$$= 491 \cdot 0.2 \cdot 3.55 \cdot 25.4$$

$$= 90.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1950 / 1 \cdot 9807 \cdot 0.2 / (11 \cdot 202713 \cdot 4.16^2 \cdot 29.5 \cdot 0.3)$$

$$= 0.587 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 660 / 1 \cdot 9807 \cdot 0.2 / (11 \cdot 188235 \cdot 4.16^2 \cdot 29.5 \cdot 0.3)$$

$$= 0.214 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1950 / 1 \cdot 9807 \cdot 0.2 / (15.8 \cdot 202713 \cdot 4.16^2 \cdot 29.5 \cdot 0.3)$$

$$= 0.408 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 660 / 1 \cdot 9807 \cdot 0.2 / (15.8 \cdot 188235 \cdot 4.16^2 \cdot 29.5 \cdot 0.3)$$

$$= 0.148 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.34, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	28.86	207.00	83.39	207.00
Radial Flange	3.36	138.00	9.70	138.00
Tangential Flange	30.51	138.00	88.17	138.00
Maximum Average	29.69	138.00	85.78	138.00
Bolting	44.07	172.00	80.86	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 44.069 mm

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 Internal Pressure Calculations: Step: 5 9:39am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS dish end		2401.8	15.062	3	188.95	137.2	1
LHS Ch Shell		2401.8	15.062	3	188.95	118	2
LHS Ch Flange		2401.9	68	3	188.95	133.2	3
Main Shell		2401.8	15.062	3	188.95	118	4
RHS Ch Flg		2401.9	64	3	202.72	134.6	5
RHS Ch Shell		2401.8	15.062	3	188.95	118	6
RHS Dish End		2401.8	10.312	3	198.45	137.2	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS dish end		2400	14749	18985	13.1794	4.64249	1
LHS Ch Shell		2400	11594	15189	13.1794	5.00876	2
LHS Ch Flange		2400	10275	10276	56	48.387	3
Main Shell		2400	11594	15189	13.1794	5.00876	4
RHS Ch Flg		2400	4588	7992	56	49.403	5
RHS Ch Shell		2400	11594	15189	13.1794	5.00876	6
RHS Dish End		2400	8345	12435	9.02335	4.72565	7

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 5004 KPa.
 Shell Side MAPnc = 10200 KPa.
 Channel Side MAWP = 4588 KPa.
 Channel Side MAPnc = 7992 KPa.

Elements Suitable for Design Internal Pressure.

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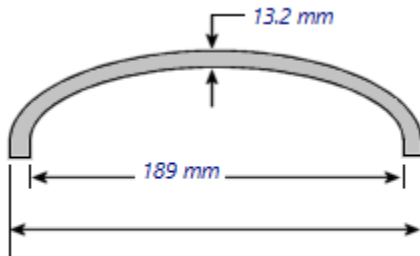
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Element: LHS dish end

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS dish end

Elliptical Head Design Information:			
Design Pressure P	2401.843 KPa.	Int. Design Temperature	270 °C
Material	SA-516 70	Ext. Design Temperature	270 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	188.951 mm	Actual Stress	22.3 N./mm ²
Spec. Min. Thk t,e, or sa	13.1794 mm	Spec. Nominal Thk.	15.062 mm
Surface Area	869.9 cm ²	SG of Contents	1.000
Element Volume	2307.9 cm ³	Weight of Contents	2.3 Kg
Empty Weight	9.6 Kg	Operating Weight	11.9 Kg
		Comp. All. Ope. Stress	123.6 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.362 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.181 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2402 \cdot 195 \cdot 0.96) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 2402)$$

$$= 1.6425 + 3.0000 = 4.6425 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.843 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 10.2) / (0.96 \cdot 195 + 0.2 \cdot 10.2)$$

$$= 14751 - 1.84 = 14749 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 13.2) / (1 \cdot 189 + 0.2 \cdot 13.2)$$

$$= 18985 \text{ KPa.}$$



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Element: LHS dish end

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2402 \cdot (0.96 \cdot 195 + 0.2 \cdot 10.2)) / (2 \cdot 1 \cdot 10.2)$$

$$= 22.339 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2402 \cdot 97.5) / (137 \cdot 1 - 0.6 \cdot 2402) + 3$$

$$= 4.725 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 1.843 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 12.1) / (97.5 + 0.6 \cdot 12.1)$$

$$= 15803 - 1.84 = 15802 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (195 / (2 \cdot 50.2)))^2 / 6$$

$$= 0.960784$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 28.489 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.2, tr = 3.12, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 15.1, tr = 3.31, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4587.820 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	13.179 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.307	Temperature Difference	84 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.362 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.181 mm

EXTERNAL PRESSURE RESULTS for: LHS dish end

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 270 °C



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Element: LHS dish end

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Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.179	215.31	21.15	0.0065664	94.80

$$MAEP = B / (K0 * Do / t) = 94.8 / (0.9 * 21.2) = 4980 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
0.580	215.31	371.11	0.0003742	34.55

$$MAEP = B / (K0 * Do / t) = 34.5 / (0.9 * 371) = 103 \text{ KPa.}$$

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Element: LHS Ch Shell

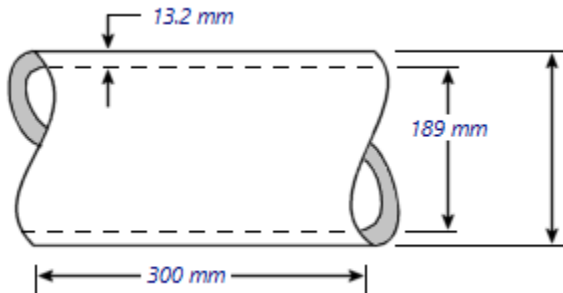
Step: 7

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	2401.843 KPa.	Int. Design Temperature	270 °C
Material	SA-106 B	Ext. Design Temperature	270 °C
Length	300.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	188.951 mm	Actual Stress	24.4 N./mm ²
Spec. Min. Thk t,e, or sa	13.1794 mm	Spec. Nominal Thk.	15.062 mm
Surface Area	2064.7 cm ²	SG of Contents	1.000
Element Volume	8413.7 cm ³	Weight of Contents	8.4 Kg
Empty Weight	22.4 Kg	Operating Weight	30.8 Kg
		Comp. All. Ope. Stress	123.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2402 \cdot 97.5) / (118 \cdot 1 - 0.6 \cdot 2402) \\
 &= 2.0088 + 3.0000 = 5.0088 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.843 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 10.2) / (97.5 + 0.6 \cdot 10.2) \\
 &= 11595 - 1.84 = 11594 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 13.2) / (94.5 + 0.6 \cdot 13.2) \\
 &= 15189 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2402 \cdot (97.5 + 0.6 \cdot 10.2)) / (1 \cdot 10.2) \\
 &= 24.442 \text{ N./mm}^2
 \end{aligned}$$



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 Element: LHS Ch Shell Step: 7 9:39am Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 13.2, tr = 1.88, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E*/(tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4587.820 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	13.179 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.185	Temperature Difference	84 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 270 °C

Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	366.55	21.15	1.7024	0.0079202	97.44

$$MAEP = (4*B)/(3*(Do/t)) = (4*97.4)/(3*21.2) = 6142 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.889	215.31	366.55	242.09	1.7024	0.0002034	18.78

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.8)/(3*242) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	19863.49	21.15	50.0000	0.0024904	80.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*80.9)/(3*21.2) = 5100 \text{ KPa.}$$



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Element: Main Shell

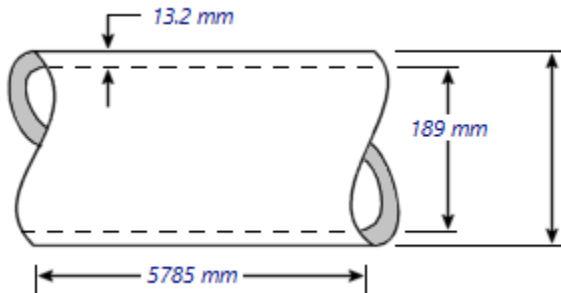
Step: 8

9:39am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	2401.843 KPa.	Int. Design Temperature	260 °C
Material	SA-106 B	Ext. Design Temperature	260 °C
Length	5785.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	188.951 mm	Actual Stress	24.4 N./mm ²
Spec. Min. Thk t,e, or sa	13.1794 mm	Spec. Nominal Thk.	15.062 mm
Surface Area	39814.9 cm ²	SG of Contents	1.000
Element Volume	102874.1 cm ³	Weight of Contents	102.8 Kg
Empty Weight	432.8 Kg	Operating Weight	535.7 Kg
		Comp. All. Ope. Stress	125.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2402 \cdot 97.5) / (118 \cdot 1 - 0.6 \cdot 2402) \\
 &= 2.0088 + 3.0000 = 5.0088 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.843 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 10.2) / (97.5 + 0.6 \cdot 10.2) \\
 &= 11595 - 1.84 = 11594 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 13.2) / (94.5 + 0.6 \cdot 13.2) \\
 &= 15189 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2402 \cdot (97.5 + 0.6 \cdot 10.2)) / (1 \cdot 10.2) \\
 &= 24.442 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell

Step: 8

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Minimum Design Metal Temperature Results:

Govrn. thk, tg = 13.2, tr = 2.05, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5003.981 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	13.179 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.201	Temperature Difference	84 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 260 °C

Elastic Modulus from Chart: CS-2 at 260 °C: 186156960 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	5785.00	21.15	26.8683	0.0025018	82.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*82.3)/(3*21.2) = 5190 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
1.949	215.31	5785.00	110.46	26.8683	0.0000920	8.57

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.57)/(3*110) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	290202.47	21.15	50.0000	0.0024904	82.27

$$MAEP = (4*B)/(3*(Do/t)) = (4*82.3)/(3*21.2) = 5186 \text{ KPa.}$$

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Element: RHS Ch Shell

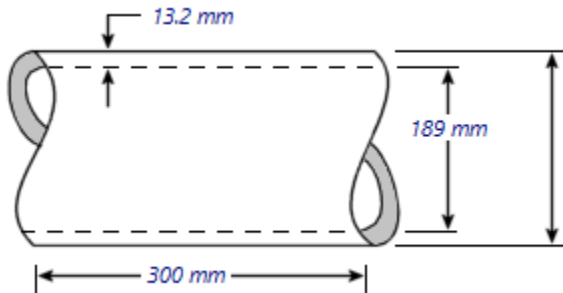
Step: 9

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	2401.843 KPa.	Int. Design Temperature	270 °C
Material	SA-106 B	Ext. Design Temperature	270 °C
Length	300.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	188.951 mm	Actual Stress	24.4 N./mm ²
Spec. Min. Thk t,e, or sa	13.1794 mm	Spec. Nominal Thk.	15.062 mm
Surface Area	2064.7 cm ²	SG of Contents	1.000
Element Volume	8413.7 cm ³	Weight of Contents	8.4 Kg
Empty Weight	22.4 Kg	Operating Weight	30.8 Kg
		Comp. All. Ope. Stress	125.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2402 \cdot 97.5) / (118 \cdot 1 - 0.6 \cdot 2402) \\
 &= 2.0088 + 3.0000 = 5.0088 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.843 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 10.2) / (97.5 + 0.6 \cdot 10.2) \\
 &= 11595 - 1.84 = 11594 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 13.2) / (94.5 + 0.6 \cdot 13.2) \\
 &= 15189 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2402 \cdot (97.5 + 0.6 \cdot 10.2)) / (1 \cdot 10.2) \\
 &= 24.442 \text{ N./mm}^2
 \end{aligned}$$



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 Element: RHS Ch Shell Step: 9 9:39am Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 13.2, tr = 1.88, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.18, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4587.820 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	13.179 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.185	Temperature Difference	84 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: RHS Ch Shell

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 270 °C

Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	367.34	21.15	1.7061	0.0079010	97.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*97.4)/(3*21.2) = 6140 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.890	215.31	367.34	241.89	1.7061	0.0002033	18.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.8)/(3*242) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.179	215.31	19902.84	21.15	50.0000	0.0024904	80.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*80.9)/(3*21.2) = 5100 \text{ KPa.}$$



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Element: RHS Dish End

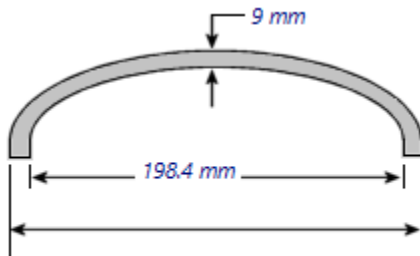
Step: 10

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:			
Design Pressure P	2401.843 KPa.	Int. Design Temperature	270 °C
Material	SA-516 70	Ext. Design Temperature	270 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	198.450 mm	Actual Stress	39.5 N./mm ²
Spec. Min. Thk t,e, or sa	9.0233 mm	Spec. Nominal Thk.	10.312 mm
Surface Area	869.9 cm ²	SG of Contents	1.000
Element Volume	2594.8 cm ³	Weight of Contents	2.6 Kg
Empty Weight	6.8 Kg	Operating Weight	9.4 Kg
		Comp. All. Ope. Stress	123.6 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.481 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.240 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2402 \cdot 204 \cdot 0.96) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 2402)$$

$$= 1.7256 + 3.0000 = 4.7256 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.843 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 6.02) / (0.96 \cdot 204 + 0.2 \cdot 6.02)$$

$$= 8347 - 1.84 = 8345 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9.02) / (1 \cdot 198 + 0.2 \cdot 9.02)$$

$$= 12435 \text{ KPa.}$$



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Element: RHS Dish End

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2402 \cdot (0.96 \cdot 204 + 0.2 \cdot 6.02)) / (2 \cdot 1 \cdot 6.02)$$

$$= 39.478 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2402 \cdot 102) / (137 \cdot 1 - 0.6 \cdot 2402) + 3$$

$$= 4.809 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 1.843 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 7.31) / (102 + 0.6 \cdot 7.31)$$

$$= 9410 - 1.84 = 9408 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (204 / (2 \cdot 52.6))^2) / 6$$

$$= 0.962528$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 19.886 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9.02, tr = 3.28, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.54$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-29 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 10.3, tr = 3.47, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.47$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-28 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-48 °C

Min Metal Temp. w/o impact per UG-20(f)

-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4587.820 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	9.023 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.545	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.481 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.240 mm

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 270 °C

Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.



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Element: RHS Dish End

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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.023	216.50	35.94	0.0038642	87.25

$$MAEP = B/(K0*Do/t) = 87.3/(0.9 *35.9) = 2697 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
0.583	216.50	371.11	0.0003743	34.55

$$MAEP = B/(K0*Do/t) = 34.5/(0.9 *371) = 103 \text{ KPa.}$$

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Shop Pressure Test Results:

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6505.176 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13260.004 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5504.380 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	5003.981 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5964.167 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	10387.897 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5046.603 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	4587.820 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS dish end	96.6	234.0	0.413	10389.75
LHS Ch Shell	105.7	216.0	0.489	10389.75
Main Shell	135.0	216.0	0.625	13261.86
RHS Ch Shell	105.7	216.0	0.489	10389.75
RHS Dish End	170.8	234.0	0.730	10389.84

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	137.20	1.006
N3-Shell inlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Pad	138.00	137.60	1.003
N4-Shell outlet	Nozzle	118.00	118.00	1.000
N4-Shell outlet	Pad	138.00	137.60	1.003
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	137.20	1.006

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS dish end	138.00	137.20	1.006
LHS Ch Shell	118.00	118.00	1.000
LHS Ch Flange	138.00	133.20	1.036
Main Shell	118.00	118.00	1.000
RHS Ch Flg	138.00	134.60	1.025
RHS Ch Shell	118.00	118.00	1.000



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Shop Pressure Test Results:

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RHS Dish End	138.00	137.20	1.006
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Minimum	1.000
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Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.30	1.001
Tubesheet Material	138.00	134.60	1.025

Minimum	1.001
---------	-------

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	53.64	216.00	0.248
N3-Shell inlet	68.47	216.00	0.317
N4-Shell outlet	68.47	216.00	0.317
N2-Tube Outlet	53.64	216.00	0.248

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 12 9:39am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6505.176 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13260.004 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5504.380 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	5003.981 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5964.167 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	10387.897 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5046.603 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	4587.820 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 2400 • 1

= 3120.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 2400 • 1

= 3120.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS dish end	29.0	234.0	0.124	3121.85
LHS Ch Shell	31.8	216.0	0.147	3121.85
Main Shell	31.8	216.0	0.147	3121.85
RHS Ch Shell	31.8	216.0	0.147	3121.85
RHS Dish End	51.3	234.0	0.219	3121.95

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	137.20	1.006
N3-Shell inlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Pad	138.00	137.60	1.003
N4-Shell outlet	Nozzle	118.00	118.00	1.000
N4-Shell outlet	Pad	138.00	137.60	1.003
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	137.20	1.006

Minimum

1.000



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Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS dish end	138.00	137.20	1.006
LHS Ch Shell	118.00	118.00	1.000
LHS Ch Flange	138.00	133.20	1.036
Main Shell	118.00	118.00	1.000
RHS Ch Flg	138.00	134.60	1.025
RHS Ch Shell	118.00	118.00	1.000
RHS Dish End	138.00	137.20	1.006

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.30	1.001
Tubesheet Material	138.00	134.60	1.025

Minimum 1.001

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	16.12	216.00	0.075
N3-Shell inlet	16.12	216.00	0.075
N4-Shell outlet	16.12	216.00	0.075
N2-Tube Outlet	16.12	216.00	0.075

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		3120.00 KPa.
Shell Thickness	ts	15.0622 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	188.951 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		3120.00 KPa.
Channel Thickness	tc	15.0622 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	188.951 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	81
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	12.7000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5860.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	15.8750 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	80.203 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	163.7 cm ²



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ASME TS Calc: Pressure Test

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Perimeter of Tube Layout Cp 476.2 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 70.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 360.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
Is the Tubesheet Flat on Tubeside? No
Step 1 Diameter on the Tubeside dt1 243.00 mm
Step 1 Depth on the Tubeside ht1 9.00 mm
Step 2 Diameter on the Tubeside dt2 192.00 mm
Step 2 Depth on the Tubeside ht2 7.00 mm
Is the Tubesheet Flat on Shellside? Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4 243.00 mm

Note: Tubesheet diameter is now: 243.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 560.000 mm
Tube End condition corresponding to Span (l) k 1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min -103.4190 KPa.
Channel Side Min. Design Pressure Ptd,min -103.4190 KPa.
Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
Flange Outside Diameter A 360.000 mm
Flange Inside Diameter B 188.951 mm
Flange Face Outside Diameter Fod 257.951 mm
Flange Face Inside Diameter Fid 188.951 mm
Gasket Outside Diameter Go 246.951 mm
Gasket Inside Diameter Gi 220.951 mm
Small end Hub thk. g0 13.1794 mm
Large end Hub thk. g1 20.0000 mm
Gasket Factor, m 3.00
Gasket Design Seating Stress y 68.95 N./mm²
Flange Facing Sketch Code Sketch 1a
Gasket Thickness tg 3.1750 mm
Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 300.000 mm



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ASME TS Calc: Pressure Test

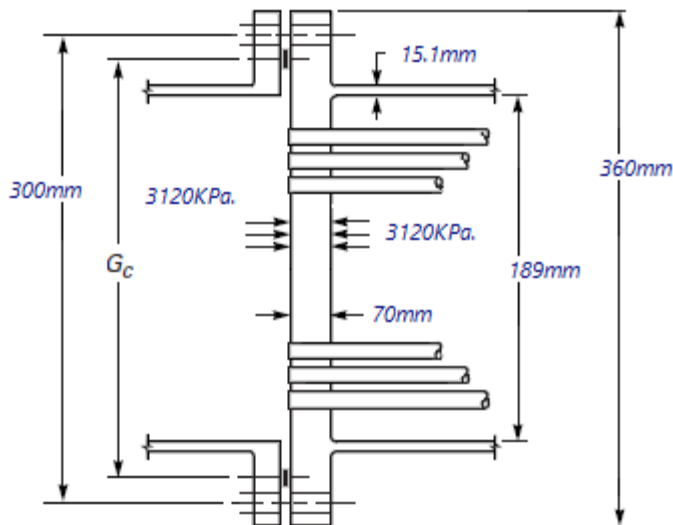
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Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	UNC Thread Series	
Number of Bolts	n	12
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 54.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= (P*(D/2-CAE)) / (S*E+0.4*P) per Appendix 1-1 (a)(1)

= (3223*(12.7/2-0))/(92.4*1+0.4*3223)

= 0.2185 + 0.0000 = 0.2185 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	12.70	5860.00	6.02	50.0000	0.0303634	95.84

EMAWP = (2.167/(D/T)-0.0833)*B = 26520 KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3968	12.70	5860.00	32.00	50.0000	0.0010740	77.38

EMAWP = (4*B)/(3*(D/T)) = (4 *77.4)/(3 *32) = 3224 KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3968 mm
Allowable Internal Pressure at Corroded thickness	35406.68 KPa.
Required Internal Design Pressure	3223.42 KPa.
Allowable External Pressure at Corroded thickness	26520.32 KPa.
Required External Design Pressure	3223.42 KPa.
Required Thickness due to Shell Side pressure	0.3968 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	N = (Goc-Gic) / 2	13.000 mm
Basic Gasket Width,	b0 = N / 2	6.500 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	6.422 mm
Gasket Reaction Diameter,	G = Go-2.0*b	234.107 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:



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Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

= l_{tx} / h (modified for corrosion if present)

= 0/70

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(12.7 - 2 \cdot 2.11 (202701232/202701232) \cdot$

$(92.4/138) \cdot (0); 12.7 - 2 \cdot 2.11)$

= 12.7000 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 80.2 + 12.7$

= 173.106 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(15.9 - 12.7) / 15.9$

= 0.200

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= 173/2

= 86.5530 mm

Effective tube pitch [p^*]:

= $p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$

= $15.9 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 173 \cdot 15.9) / (\pi \cdot 173^2) }$

= 15.8750 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(15.9 - 12.7) / 15.9$

= 0.2000

Ratio [ρ_s]:

= a_s / a_o

= 94.5/86.6

= 1.0915

Ratio [ρ_c]:

= a_c / a_o

= 117/86.6

= 1.3524

Compute parameter [xt]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$

= $1 - 81 ((12.7 - 2 \cdot 2.11) / (2 \cdot 86.6))^2$

= 0.8056

Determine parameter [xs]:

= $1 - Nt(dt / (2 \cdot a_o))^2$

= $1 - 81 (12.7 / (2 \cdot 86.6))^2$

= 0.5640

Determine the value [$h'g$]:

= $\max((h_g - CATC); 0)$ (For pressure only cases)

= $\max((0 - 0); 0)$

= 0.000 mm



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ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 15.1 (189 + 15.1) 202701232 / 5860$$

$$= 34052.6562 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 202701232 / 5860$$

$$= 247.6188 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 34053 / (81 \cdot 248)$$

$$= 1.69778$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 34053/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((189 + 15.1)15.1)$$

$$= 0.0328 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.033 \cdot 202701232 \cdot 15.1^3 / (6(1 - 0.3^2))$$

$$= 4160175104.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(189 + 15.1) \cdot 4.16E+09 / 70^3 \cdot (1 + 70 \cdot 0.033 + (70 \cdot 0.033)^2 / 2)$$

$$= 88043448.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 189^2 / (4 \cdot 202701232 \cdot 15.1) \cdot (1 - 189 \cdot 0.3 / (2(189 + 15.1)))$$

$$= 0.25171E-02 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 4.409449$; $\mu^* = 0.200000$; $E^* = 31489462 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (4.409) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2)81 \cdot 202701232 \cdot 2.11$$

$$(12.7 - 2.11)86.6^2) / (31489462 \cdot 5860 \cdot 70^3))^{1/4}$$

$$= 0.9525$$

Values from Table 4.18.3

$Z_d = 2.482893$; $Z_v = 0.170039$; $Z_m = 0.681610$
 $Z_a = 0.350844E+00$; $Z_w = 0.170039$

ASME Section VIII-2 Part 4.18.8.4 Step 4:



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Diameter ratio [K]:

$$= A / D_o$$

$$= 243/173$$

$$= 1.4038$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31489462 \cdot (88043448 + 0 + 202701232 \cdot \ln(1.4))$$

$$= 2.6959$$

Compute parameter [\phi]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.46) 2.7$$

$$= 3.9321$$

Compute parameter [Q1]:

$$= (\psi s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.09 - 1 - 3.93 \cdot 0.17) / (1 + 3.93 \cdot 0.68)$$

$$= -0.156808$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (2.48 + -0.16 \cdot 0.17) / 2 \cdot 0.95^4$$

$$= 1.0108$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17 + -0.16 \cdot 0.68) / 2 \cdot 0.95^4$$

$$= 0.0260$$

Compute parameter [U]:

$$= (Z_w + (\psi s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.17 + (1.09 - 1) 0.68) 0.95^4 / (1 + 3.93 \cdot 0.68)$$

$$= 0.0520$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [\gamma_b]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (234 - 300) / 173 = -0.38$$

Compute Parameter [\gamma]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [\omega s]:

$$= \psi s \cdot k_s \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.09 \cdot 4.16E+09 \cdot 0.033 \cdot 0.00251714 (1 + 70 \cdot 0.033)$$

$$= 1235.3409 \text{ mm}^2$$

Calculate parameter [\omega*s]:

$$= a o^2 (\psi s^2 - 1) \cdot (\psi s - 1) / 4 - \omega s$$

$$= 86.6^2 (1.09^2 - 1) \cdot (1.09 - 1) / 4 - 1235$$

$$= -1202.5233 \text{ mm}^2$$

Parameter [\omega c]:

$$= \rho c \cdot k c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.35 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 70) - 0.000 (0/70 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [\omega*c]:

$$= a o^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\psi s - 1) / 2] - \omega c$$

$$= 86.6^2 [(1.35^2 + 1) \cdot (1.35 - 1) / 4 - (1.09 - 1) / 2] - 0$$



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$$= 1524.1832 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned} &= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - \\ &\quad [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \} \\ &= 3120 \{ 0.56 + 2(1 - 0.56) 0.3 + [2/1.7 \cdot \\ &\quad (189/173)^2] 0.3 - [(1.09^2 - 1)/(1 \cdot 1.7)] - \\ &\quad [(1 - 1)/(2 \cdot 1 \cdot 1.7)] [(0^2 - (189)^2)/173^2] \} \\ &= 3537.7852 \text{ KPa.} \end{aligned}$$

Pressure [P't]:

$$\begin{aligned} &= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t \\ &= [0.81 + 2(1 - 0.81) 0.3 + 1/(1 \cdot 1.7)] 3120 \\ &= 4715.1025 \text{ KPa.} \end{aligned}$$

Pressure [P_y]:

$$\begin{aligned} &= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\ &= 81 \cdot 248 \cdot 0 / (3.14 \cdot 86.6^2) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} &= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\ &= -0.38 \cdot 0.052 \cdot 74816 / (2 \cdot \pi \cdot 86.6^2) \\ &= 308.3961 \text{ KPa.} \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned} &= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t) \\ &= -(0.052/86.6^2) \cdot (-1203 \cdot 3120 - 1524 \cdot 3120) \\ &= 59.0277 \text{ KPa.} \end{aligned}$$

Calculate the pressure [P_w]:

$$\begin{aligned} &= (U/a_o^2) \cdot (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c) \\ &= (0.052/86.6^2) \cdot (1235 \cdot 0 - 0 \cdot 0) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (p_s - 1) Qz2)) \cdot \\ &\quad (P^* s - P^* t + P_y + P_w + P_w + Prim) \\ &= 0.1000E+01 \cdot 1.7 / (1 + 1 \cdot 1.7 (1.01 + (1.09 - \\ &\quad 1) 0.026)) \cdot (3538 - 4715 + 0 + 0 + 308 + 59) \\ &= -505.5082 \text{ KPa.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c)) + \\ &\quad W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-1203 \cdot 3120 - 1524 \cdot 3120) - \\ &\quad (1235 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\ &\quad 74816 \cdot -0.38 / (2 \cdot \pi)] / (1 + 3.93 \cdot 0.68) \\ &= -1467.3518 \text{ Kg} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2) \\ &= -0.16 + 2 \cdot -1467 / (-506 \cdot 86.6^2) \\ &= 7.44252 \end{aligned}$$



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Fm Value from Table 4.18.3 = 3.881648

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 3.88 / 0.2) \cdot (2 \cdot 86.6 / (70 - 0))^2 \cdot -506$$

$$= -90.0034 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta_T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.31 / 0.2) \cdot (2 \cdot 86.6 / (11.9 - 0))^2 \cdot 734$$

$$= 356.2436 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 11.9 + 0 + 0$$

$$= 11.8925 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta_T \cdot S_y)$$

$$= (2 \cdot 0.2 \cdot 70 / 86.6) \cdot 190$$

$$= 61.4641 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 14.0613 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(11.9; 3.81)$$

$$= 11.8925 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.45, F_{tmax} = 1.54$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((3120 \cdot 0.56 - 3120 \cdot 0.81) - (-506) \cdot 0.45) / (0.81 - 0.56)$$

$$= -2.1683 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((3120 \cdot 0.56 - 3120 \cdot 0.81) - (-506) \cdot 1.54) / (0.81 - 0.56)$$

$$= 0.1074 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.17 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 170.9969 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:



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$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(2.48 + 7.44 \cdot 0.17)0.95^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (12.7^2 + (12.7 - 2 \cdot 2.11)^2) }) / 4$$

$$= 3.8177 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 202701232 / 180) }$$

$$= 159.3101$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 560 / 3.82$$

$$= 146.6842$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (180 / 2) \cdot (1 - 147 / (2 \cdot 159))$$

$$= 170.997 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.17 \cdot 0.7$$

$$= 15.52 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (12.7 - 2.11) 2.11$$

$$= 0.702 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 171$$

$$= 1224.018 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (12.7 + 0.67 \cdot 2.11) 171$$

$$= 1224.0182 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171 / 171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (12.7 + 0.67 \cdot 2.11) 171$$

$$= 897.2130 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o })$$

$$= 2 (\sqrt{ ((0.75 \cdot 12.7)^2 + 1.07 \cdot 2.11 \cdot (12.7 - 2.11) 1 \cdot 1) - 0.75 \cdot 12.7 })$$



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= 2.364 mm

Minimum Required Fillet Weld Leg [afr]: 1.2660 mm

Minimum Required Groove Weld Leg [agr]: 1.2660 mm

Tube-Tubesheet Jt allowable, 1224 is >= tube strength 1224 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 240 = 342 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 86.6^2 / ((189 + 15.1) 15.1) \cdot [-506 + (1.09^2 - 1)(3120 - 3120)] + 94.5^2 \cdot 3120 / ((189 + 15.1) 15.1)$$

$$= 7.8305 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 15.1 mm, the minimum Shell length adjacent to the tubesheet should be 96.026 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	90.0 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	1.2 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	2.2 <=	171.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.2 <=	-171.0 N./mm ²	Pass
Maximum Force on any one Tube	15.5 <=	1224.0 Kgf	Pass
Axial Membrane Stress in Shell	7.8 <=	342.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	11.893	70.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.266	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.266	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	3.810 28.742	98	356	7 190	Ps+Pt-Th D1	Ok
D2uc	8.773 ...	14	356	6 190	Ps+Pt-Th D2	Ok
D3uc	11.893 ...	90	356	1 190	Ps+Pt-Th D3	Ok
Max:	11.8925 28.742 mm	0.275	0.036 (Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures



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derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Shell Stresses				:	Shell Band Stress				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	1	341	...	...	:	...	...	...	...	:	Ok
D2uc	7	341	...	...	:	...	...	...	...	:	Ok
D3uc	8	341	...	...	:	...	...	...	...	:	Ok
Max RATIO	0.023				:	...				:	...

Tube, Shell and Channel Stress Summary:

Case#	Tube Stresses				Tube Loads		Shell Stress		Channel Stress	Stress Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd Fail
D1uc	2	170	...	-170	15	1224	...	341	...	Ok
D2uc	2	170	-2	-170	14	1224	...	341	...	Ok
D3uc	2	170	-2	-170	16	1224	...	341	...	Ok
Max RATIO	0.013		0.013		0.013		...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	11.893	70.000	Ok
Tubesheet thickness flanged extension :	28.742	54.000	Ok
Tube thickness :	0.397	2.110	Ok
Tube-tubesheet fillet weld leg :	1.266	2.110	Ok
Tube-tubesheet groove weld leg :	1.266	2.110	Ok

Min Shell length of thk, (15.062) adj. to tubesheet: 96.026 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.081 mm
 Due to Pressure + Thermal Shell Compresses by : -0.081 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded



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1	0.0074	...	...	...	...	...
2	-0.0815	...	...	...	...	...
3	-0.0767	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 13 9:39am Sep 9,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	215.31	10.1794	0.0065664	94.8001
20	30	366.546	215.309	10.1794	0.0079202	97.442
30	50	No Calc	...	53	No Calc	No Calc
50	60	5785	215.309	10.1794	0.0025018	82.3383
60	80	No Calc	...	53	No Calc	No Calc
80	90	367.337	215.309	10.1794	0.007901	97.4078
90	100	No Calc	216.497	6.02335	0.0038642	87.2503

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	13.1794	3.58017	103.419	4980
20	30	13.1794	3.88937	103.419	6142
30	50	56	48.387	103.419	No Calc
50	60	13.1794	4.94924	103.419	5190
60	80	56	49.403	103.419	No Calc
80	90	13.1794	3.89012	103.419	6140
90	100	9.02335	3.58337	103.419	2697

Minimum

2697

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	366.546	19863.5	No Calc	No Calc
30	50	No Calc	No Calc	No Calc	No Calc
50	60	5785	290202	No Calc	No Calc
60	80	No Calc	No Calc	No Calc	No Calc
80	90	367.337	19902.8	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 14

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	9.61628	2307.93	7.70096	2486.69	1.92326
20	30	22.4464	8413.67	18.24	8956.5	4.48928
30	50	41.0259	2759.34	39.7078	2806.38	8.20517
50	60	432.841	102874	351.728	113342	86.5683
60	80	35.2725	2706.59	34.0704	2737.62	7.05451
80	90	22.4464	8413.67	18.24	8956.5	4.48928
90	100	6.83193	2594.79	4.84444	2786.9	1.36639
Total		570	130070.11	474	142072.16	114



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	2.30354	-15.7459	1.9755	...	Liquid: 10
10	Insl	5.17533	25.4	...	...	Insulation
20	Liqd	8.36565	150	1.97531	...	Liquid: 20
20	Insl	10.4727	150	...	...	Insulation
20	Nozl	28.7778	150	138.975	...	N1-Tube Inlet
30	Liqd	2.75765	41	-3.02451	...	Liquid: 30
30	Insl	6.5431	41.5	...	...	Insulation
50	Sadl	95.0115	1200	301.148	...	LH floating sad
50	Sadl	95.0115	4650	301.148	...	RH fixed saddle
50	Liqd	102.811	2892.5	1.97531	...	Liquid: 50
50	Insl	201.774	2890	...	...	Insulation
50	Nozl	28.6208	600	138.975	...	N3-Shell inlet
50	Nozl	28.6208	5250	-138.975	...	N4-Shell outlet
60	Liqd	2.67123	36	3.85849	...	Liquid: 60
60	Insl	5.50906	41.5	...	...	Insulation
80	Liqd	8.36565	150	1.97531	...	Liquid: 80
80	Insl	10.4727	150	...	...	Insulation
80	Nozl	28.6208	150	-138.975	...	N2-Tube Outlet
90	Liqd	2.55909	67.3375	6.725	...	Liquid: 90
90	Insl	5.25174	25.4	...	...	Insulation
30	FTsh	59.5878	120.175	...	...	Fixed Tubesheet
30	Tube	317.304	3085.18	...	...	Tubes
30	RTsh	59.5878	6050.17	...	...	Tubesheet
50	Baf1	5.59703	560	...	...	Baffles
50	Baf1	5.59703	1120	...	...	Baffles
50	Baf1	5.59703	1680	...	...	Baffles
50	Baf1	5.59703	2240	...	...	Baffles
50	Baf1	5.59703	2800	...	...	Baffles
50	Baf1	5.59703	3360	...	...	Baffles
50	Baf1	5.59703	3920	...	...	Baffles
50	Baf1	5.59703	4480	...	...	Baffles
50	Baf1	5.59703	5040	...	...	Baffles
50	Baf1	5.66053	5600	...	...	Baffles

Total Weight of Each Detail Type:



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Element and Detail Weights: Step: 14 9:39am Sep 9,2026

Saddles	190.0
Liquid	129.8
Insulation	245.2
Nozzles	114.6
Exchanger Components	492.5
Liquid in Tubes	27.4

Sum of the Detail Weights 1199.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	684.6	684.6	684.6	684.6	684.6	684.6
Saddles	190.0	190.0	190.0	190.0	190.0	190.0
Nozzles	114.6	114.6	114.6	114.6	114.6	114.6
Exchanger	492.5	492.5	492.5	492.5	492.5	492.5
Insulation	...	...	...	...	245.2	245.2 *
Ope. Liquid	...	...	...	...	...	129.8
Tube Ope Lqd	...	...	...	...	...	27.4
Test Liquid	...	130.0	...	...	...	...
Tube Tst Lqd	...	27.4	...	...	...	...
Totals	1481.8	1639.2	1481.8	1481.8	1727.0	1884.2

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 20.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1481.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	1639.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1481.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1481.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1727.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1884.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	1769.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	1781.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	27453.4 cm ³
Weight of Ope Liq in tubes :	27.4 kg.
Weight of Water in tubes :	27.4 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	869.877	688.562
20	30	2064.73	1780.82
30	50	1693.54	557.991
50	60	39814.9	34340.1
60	80	1532.91	509.483
80	90	2064.73	1780.82
90	100	869.873	743.612



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Element and Detail Weights:

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Total	48910.6	40401.4 cm ²
Total	52.6	43.5 ft ²

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Nozzle Flange MAWP:

Step: 15 9:39am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	8226.1	10200.0	270	600	GR 1.1	7458.2	4881.0	8226.1	4881.0
N3-Shell inlet	8308.1	10200.0	260	600	GR 1.1	7458.2	5004.0	8308.1	5004.0
N4-Shell outlet	8308.1	10200.0	260	600	GR 1.1	7458.2	5004.0	8308.1	5004.0
N2-Tube Outlet	8226.1	10200.0	270	600	GR 1.1	7458.2	4881.0	8226.1	4881.0

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 5003.981 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 10200.003 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 4880.982 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10200.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	111.125	0.50	185	734	4587.8	4881.0	Pass
N3-Shell inlet	111.125	0.50	185	734	5004.0	5004.0	Pass
N4-Shell outlet	111.125	0.50	185	734	5004.0	5004.0	Pass
N2-Tube Outlet	111.125	0.50	185	734	4587.8	4881.0	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(185) + 4 \cdot \text{abs}(734) \cdot 111.125 \leq \pi \cdot 111.125^3 [(8226.086 - 4587.820) + .500 \cdot 8226.086]$$

$$= 0.32788E+04 \leq 0.34077E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8226.086, 0.500 \cdot 8226.086 + 8226.086 - 3278.8 / (\pi \cdot 111.125^3))$$

$$= \min(8226.086, 4880.982)$$

$$= 4880.982 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(185) + 4 \cdot \text{abs}(734) \cdot 111.125 \leq \pi \cdot 111.125^3 [(8308.086 - 5003.981) + .500 \cdot 8308.086]$$

$$= 0.32788E+04 \leq 0.32788E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8308.086, 0.500 \cdot 8308.086 + 8308.086 - 3278.8 / (\pi \cdot 111.125^3))$$

$$= \min(8308.086, 5003.981)$$

$$= 5003.981 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:



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Nozzle Flange MAWP:

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$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(185) + 4 \cdot \text{abs}(734) 111.125 \leq \pi \cdot 111.125^3 [(8308.086 - 5003.981) + .500 \cdot 8308.086] \\ &= 0.32788E+04 \leq 0.32788E+04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8308.086, 0.500 \cdot 8308.086 + 8308.086 - 3278.8 / (\pi \cdot 111.125^3)) \\ &= \min(8308.086, 5003.981) \\ &= 5003.981 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(185) + 4 \cdot \text{abs}(734) 111.125 \leq \pi \cdot 111.125^3 [(8226.086 - 4587.820) + .500 \cdot 8226.086] \\ &= 0.32788E+04 \leq 0.34077E+04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8226.086, 0.500 \cdot 8226.086 + 8226.086 - 3278.8 / (\pi \cdot 111.125^3)) \\ &= \min(8226.086, 4880.982) \\ &= 4880.982 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.99) / (1 + 1.7 \cdot 3.4 \cdot 0.23))) \\
 &= \min(0.85, 0.92) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.657
Structure Height to Diameter ratio	17.193

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 535 + 300 = 835 \text{ mm} \\
 &= 2.74 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (2.74 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900)^{2/9.5} \\
 &= 0.85
 \end{aligned}$$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.66 • 0.84

= 7.46 lbs. [3.39] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS dish end	835.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Shell	835.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Flange	835.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	835.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Flg	835.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Shell	835.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Dish End	835.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	835	498.372	776.427	78.12	3.72389
20	30	835	498.371	1495.11	78.12	7.17084
30	50	835	466.741	382.728	78.12	1.83563
50	60	835	498.164	28818.8	78.12	138.22
60	80	835	483.26	347.948	78.12	1.66882
80	90	835	498.371	1495.11	78.12	7.17084
90	100	835	499.796	772.819	78.12	3.70658

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation:

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(2.090.75)$
 $= 0.035$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.035, 0.03)$ per 12.8.2
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 1694 \cdot 1.25$
 $= 99.619$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 1694$
 $= 53.1$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	94.4755	188.244	10.9194



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Earthquake Load Calculation:

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20	30	94.4753	188.244	10.9194
30	50	94.4755	188.244	10.9194
50	Sad1	94.4753	188.244	10.9194
Sad1	60	94.4753	188.244	10.9194
50	60	94.4753	188.244	10.9194
60	80	101.358	188.244	11.7149
80	90	94.4753	188.244	10.9194
90	100	99.225	188.244	11.4683

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:39am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3430.975 mm
Center of Gravity of the Liquid	3402.332 mm
Center of Gravity of the Insulation	3384.257 mm
Center of Gravity of the Nozzles	3407.828 mm
Center of Gravity of the Tubesheet(s)	3435.975 mm
Center of Gravity of the Tubes	3435.975 mm
Center of Gravity of the Baffles	3588.831 mm
Center of Gravity of Bare Shell New and Cold	3350.823 mm
Center of Gravity of Bare Shell Corroded	3341.542 mm
Vessel CG in the Operating Condition	3398.422 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3399.596 mm
Vessel CG in the Test Condition	3400.540 mm

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Saddle Calcs: Road Transport:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Note:

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

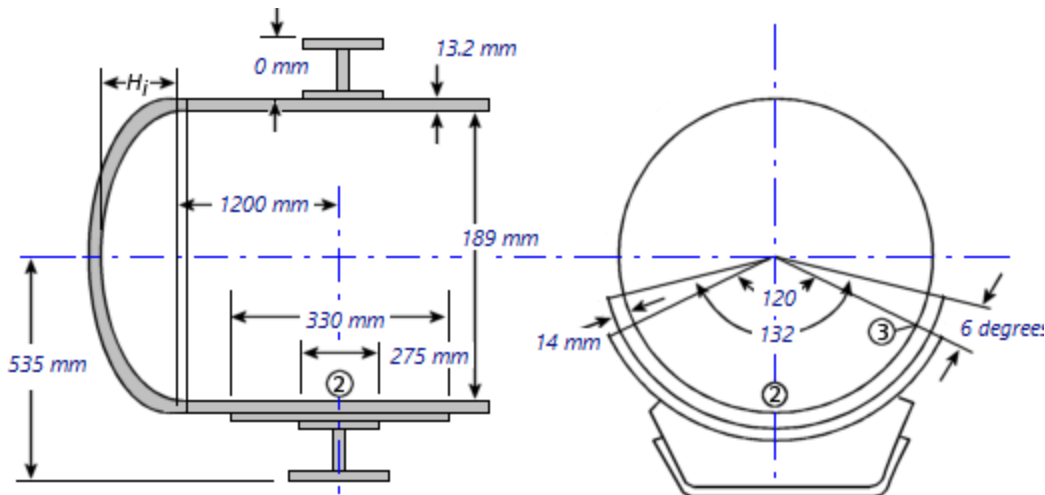
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{101 \cdot 13.2}, 2 \cdot 1200)$$

$$= 331.9342 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	101.07	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		133.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		1053.62	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	97.2	Km/hr



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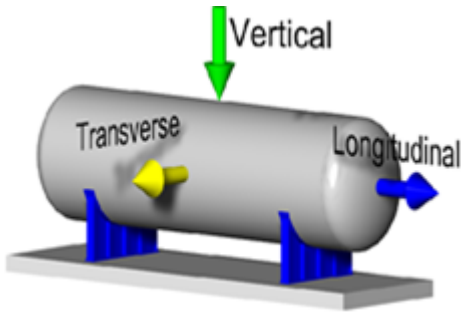
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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-6.04	-216.00
Long. Stress at Top of Midspan	6.04	118.00
Long. Stress at Bottom of Midspan	6.04	118.00
Long. Stress at Top of Saddles	56.84	118.00
Long. Stress at Bottom of Saddles	-31.51	-216.00
Long. Stress at Bottom of Saddles	31.51	118.00
Tangential Shear in Shell	5.31	94.40
Circ. Stress at Horn of Saddle	5.31	177.00
Circ. Compressive Stress in Shell	0.18	118.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 36.7 \cdot 0.00256 \cdot 60.4^2 \cdot 1 \\
 &= 342.7 \text{ lbf} \\
 &= 155.437 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 1.36^2/4 \cdot 0.00256 \cdot 60.4^2 \cdot 1 \cdot 1.2 \\
 &= 16.3 \text{ lbf} \\
 &= 7.4 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 1292 \cdot 1 + 155 \\
 &= 284.6 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 1292 \cdot 1 + 7.41 \\
 &= 136.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 285/2 \cdot 535/211 \\
 &= 361.3 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 137 \cdot 535/3450 \\
 &= 21.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 659 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 19 9:39am Sep 9, 2026

= 32.967 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 659 + 33 + max(21.2, 361)

= 1053.6 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1148.64	Kgf
Transverse Shear Load	142.31	Kgf
Longitudinal Shear Load	136.59	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(189/2 + 13.2 + 14)sin(120/2), 2•101)

= 210.712 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -1054•1200[1-(1-1200/5785+(101²-0²)/

(2•1200•5785))/(1+(4•0)/(3•5785))]

= -261.3 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 1054•5785/4((1+2(101²-0²)/(5785²))/(1+(4•0)/

(3•5785))-4•1200/5785)

= 260.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 101/(2•13.2) - 260/(π•101²•13.2)

= -6.04 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 101/(2 • 13.2) + 260/(π • 101² • 13.2)

= 6.04 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•101/(2•13.2)--261/(0.11•π•101²•13.2)

= 56.84 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•101/(2•13.2)+-261/(0.19•π•101²•13.2)

= -31.51 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 1054 (5785 - 2 • 1200)/(5785 + (4 • 0/3))

= 616.5 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ2]:

= K2 • T / (Rm • t)



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$$= 1.17 \cdot 617 / (101 \cdot 13.2)$$

$$= 5.31 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(101 \cdot 13.2)}$$

$$= 28.467 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 1054 \cdot 0.1 / (13.2 (275 + 28.5 + 28.5))$$

$$= -0.18 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -1054 / (4 \cdot 13.2 (275 + 28.5 + 28.5)) -$$

$$3 \cdot 0.053 \cdot 1054 / (2 \cdot 13.2^2)$$

$$= -5.31 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(101 \cdot 13.2)}, 2 \cdot 1200)$$

$$= 331.93 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	440.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(94.5 \cdot 13.2)} + 330)13.2^3) / 12$$

$$= 7.345 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 7.35 + 50.7 (165 - 6.59)^2$$

$$= 12747.072 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	385.0	13.2	6.6	50.7	33440.8	0.127E+05
Wearplate	330.0	14.0	20.2	46.2	93228.9	0.970E+04
Web	14.0	399.3	226.9	55.9	1268292.2	0.957E+04
BasePlate	300.0	14.0	433.5	42.0	1820803.8	0.303E+05
Totals	...	...	...	194.9	3215765.8	0.623E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3215765 / 195$$

$$= 165.034 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$



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$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1054$$

$$= 214.4353 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1459 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 535 - 94.5 \cdot \sin(120/2) / 1.05$$

$$= 456.870 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 97.9709 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2545 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1054 + 95)300) / (4 \cdot 600 \cdot 173)}$$

$$= 4.937 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$$

$$= 210.712 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 211 / (3 - 1)$$

$$= 105.356 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 105 \cdot 300 / 2$$

$$= 158.034 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1149 / (600 \cdot 300)$$

$$= 0.638 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 158 \cdot 0.64$$

$$= 100.846 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 7.37$$

$$= 43.915 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 101 / 43.9$$



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= 0.225 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 275 - 14

= 261.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/43.9

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 275/2

= 137.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(2426/43.9)}$

= 74.330 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 535 - cos(120/2)(94.5 + 13.2 + 14) - 14

= 460.173 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 460/74.3

= 6.191

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 137 / (2 • 600) • 105 • 460/2

= 2.759 Kg-m.

Compressive Allowable, KL/r < Cc (6.19 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.19)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (6.19) / (8 \cdot 123) - (6.19^3) / (8 \cdot 123^3))$

= 154.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 0.23/154 + (2.76 • 138/24263022) / 173

= 0.002

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
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Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/51.3

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 275/2

= 137.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 535 - cos(122/2)(14 + 0 + 0) - 0

= 399.345 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(2428/51.3)}$

= 68.809 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 399/68.8

= 5.804

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= 137/(2 • 600)

= 0.114 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.11 • 105 • 399

= 4.789 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 4.79/176610

= 0.266 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.8 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (5.8)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (5.8) / (8 \cdot 123) - (5.8^3) / (8 \cdot 123^3))$

= 154.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$

= 0.39/154 + (4.79 • 138/2428)/173

= 0.004

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$

= $\min(275 + 1.56 \cdot \sqrt{101 \cdot 13.2}, 2 \cdot 1200)$



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= 331.9342 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	101.07	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		134.60	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		1025.32	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	97.2	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-5.88	-216.00
Long. Stress at Top of Midspan	5.88	118.00
Long. Stress at Bottom of Midspan	5.88	118.00
Long. Stress at Top of Saddles	55.32	118.00
Long. Stress at Bottom of Saddles	-30.66	-216.00
Long. Stress at Bottom of Saddles	30.66	118.00
Tangential Shear in Shell	5.17	94.40
Circ. Stress at Horn of Saddle	5.16	177.00
Circ. Compressive Stress in Shell	0.17	118.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 36.7 \cdot 0.00256 \cdot 60.4^2 \cdot 1$$

$$= 342.7 \text{ lbf}$$

$$= 155.437 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 1.36^2/4 \cdot 0.00256 \cdot 60.4^2 \cdot 1 \cdot 1.2$$

$$= 16.3 \text{ lbf}$$

$$= 7.4 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 1292 \cdot 1 + 155$$

$$= 284.6 \text{ Kgf}$$



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Longitudinal Force due to Axial Acceleration [FI]:

$$= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$$

$$= 0.1 \cdot 1292 \cdot 1 + 7.41$$

$$= 136.6 \text{ Kg}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$= F_t / \text{Num of Saddles} \cdot B / E$$

$$= 285/2 \cdot 535/211$$

$$= 361.3 \text{ Kg}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$= F_l \cdot B / L_s$$

$$= 137 \cdot 535/3450$$

$$= 21.2 \text{ Kg}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$= \text{Saddle Load} \cdot g_y$$

$$= 632 \cdot 0.05$$

$$= 31.619 \text{ Kg}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + F_{rv} + \max(F_{wl}, F_{wt})$$

$$= 632 + 31.6 + \max(21.2, 361)$$

$$= 1025.3 \text{ Kg}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1120.33	Kgf
Transverse Shear Load	142.31	Kgf
Longitudinal Shear Load	136.59	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(189/2 + 13.2 + 14) \sin(120/2), 2 \cdot 101)$$

$$= 210.712 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1025 \cdot 1200 [1 - (1 - 1200/5785 + (101^2 - 0^2)/(2 \cdot 1200 \cdot 5785))/(1 + (4 \cdot 0)/(3 \cdot 5785))]$$

$$= -254.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 1025 \cdot 5785/4 [(1 + 2(101^2 - 0^2)/(5785^2))/(1 + (4 \cdot 0)/(3 \cdot 5785)) - 4 \cdot 1200/5785]$$

$$= 253.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 0 \cdot 101/(2 \cdot 13.2) - 253/(\pi \cdot 101^2 \cdot 13.2)$$

$$= -5.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 101/(2 \cdot 13.2) + 253/(\pi \cdot 101^2 \cdot 13.2)$$

$$= 5.88 \text{ N./mm}^2$$



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Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 101 / (2 \cdot 13.2) - 254 / (0.11 \cdot \pi \cdot 101^2 \cdot 13.2)$$

$$= 55.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 101 / (2 \cdot 13.2) + 254 / (0.19 \cdot \pi \cdot 101^2 \cdot 13.2)$$

$$= -30.66 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1025 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3))$$

$$= 600.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 600 / (101 \cdot 13.2)$$

$$= 5.17 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{101 \cdot 13.2}$$

$$= 28.467 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 1025 \cdot 0.1 / (13.2 (275 + 28.5 + 28.5))$$

$$= -0.17 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -1025 / (4 \cdot 13.2 (275 + 28.5 + 28.5)) - 3 \cdot 0.053 \cdot 1025 / (2 \cdot 13.2^2)$$

$$= -5.16 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{101 \cdot 13.2}, 2 \cdot 1200)$$

$$= 331.93 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 \cdot \sqrt{94.5 \cdot 13.2}) + 330)^3 / 12$$

$$= 7.345 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 7.35 + 50.7 (165 - 6.59)^2$$



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= 12747.072 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	385.0	13.2	6.6	50.7	33440.8	0.127E+05
Wearplate	330.0	14.0	20.2	46.2	93228.9	0.970E+04
Web	14.0	399.3	226.9	55.9	1268292.2	0.957E+04
BasePlate	300.0	14.0	433.5	42.0	1820803.8	0.303E+05
Totals	...	...	...	194.9	3215765.8	0.623E+05

Distance to Centroid [C1]:

= AY / A

= 3215765/195

= 165.034 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.2 • 1025

= 208.6752 Kgf

Tension Stress, St = (Fh/As) = 0.1420 N./mm²

Allowed Stress, Sa = 0.6 • Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 535 - 94.5 • sin(120/2)/1.05

= 456.870 mm

Bending Moment, M = Fh • d = 95.3392 Kg-m.

Bending Stress, Sb = (M • C1 / I) = 0.2477 N./mm²

Allowed Stress, Sa = 2/3 • Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(1025 + 95)300/(4 • 600 • 173))

= 4.875 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120/2) (94.5 + 13.2 + 14)

= 210.712 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 211/(3 - 1)

= 105.356 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 105 • 300/2

= 158.034 cm²



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Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1120 / (600 \cdot 300)$$

$$= 0.622 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 158 \cdot 0.62$$

$$= 98.362 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 7.37$$

$$= 43.915 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 98.4 / 43.9$$

$$= 0.220 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 43.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275 / 2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2426 / 43.9)}$$

$$= 74.330 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 535 - \cos(120/2) (94.5 + 13.2 + 14) - 14$$

$$= 460.173 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 460 / 74.3$$

$$= 6.191$$

Bending Moment [Rm]:



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$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 137 / (2 \cdot 600) \cdot 105 \cdot 460 / 2$$

$$= 2.759 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (6.19 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.19)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.19) / (8 \cdot 123) - (6.19^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.22 / 154 + (2.76 \cdot 138 / 24263022) / 173$$

$$= 0.002$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 51.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275 / 2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 535 - \cos(122/2) (14 + 0 + 0) - 0$$

$$= 399.345 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2428 / 51.3) }$$

$$= 68.809 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 399 / 68.8$$

$$= 5.804$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 137 / (2 \cdot 600)$$

$$= 0.114 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.11 \cdot 105 \cdot 399$$

$$= 4.789 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.79 / 176610$$

$$= 0.266 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.8 < 123) per AISC E2-1 [Sca]:



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$$\begin{aligned}
 &= (1 - (K_{lr})^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_{lr}) / (8 \cdot C_c) - (K_{lr}^3) / (8 \cdot C_c^3)) \\
 &= (1 - (5.8)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (5.8) / (8 \cdot 123) - (5.8^3) / (8 \cdot 123^3)) \\
 &= 154.2 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\
 &= 0.38 / 154 + (4.79 \cdot 138 / 2428) / 173 \\
 &= 0.004
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Operating Case:

Step: 20

9:39am

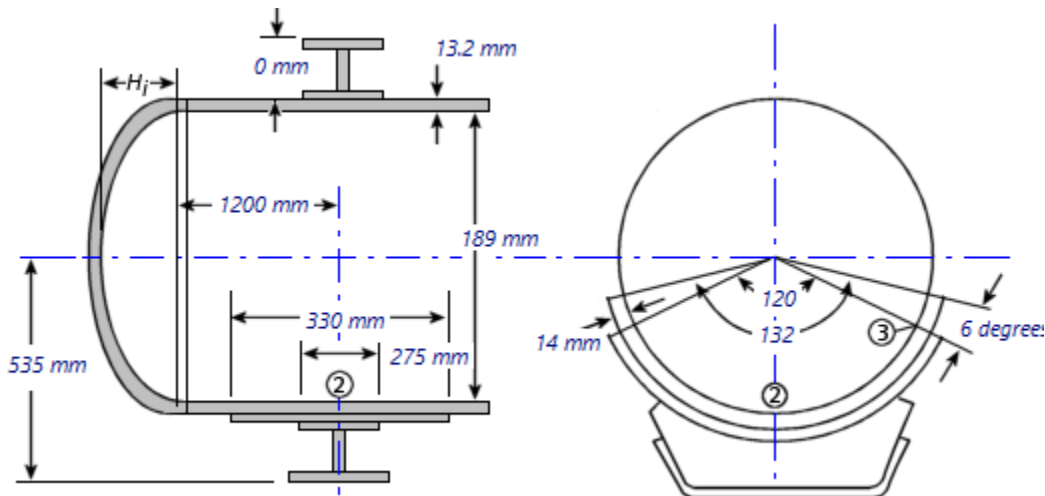
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		133.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	1514.18	Kgf
Pressure used in Analysis	P	2400.887 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	1.19	118.00
Long. Stress at Bottom of Midspan	23.01	118.00
Long. Stress at Top of Saddles	114.78	118.00
Long. Stress at Bottom of Saddles	-44.82	-176.76
Long. Stress at Bottom of Saddles	44.82	118.00
Tangential Shear in Shell	9.74	94.40
Circ. Stress at Horn of Saddle	2.49	147.50



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 Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Circ. Compressive Stress in Shell	0.14	118.00
-----------------------------------	------	--------

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (163/2 + 0) \cdot 535/211$
 $= 622.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(16.8, 0) \cdot 535/3450$
 $= 2.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(99.6, 0) \cdot 535/3450$
 $= 15.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (99.6/2 + 0) \cdot 535/211$
 $= 379.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 892 + \max(2.6, 623, 15.4, 379)$
 $= 1514.2 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (0.42 \cdot 1.2)^2)$
 $= 164.386 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1609.19	Kgf
Transverse Shear Load	81.75	Kgf
Longitudinal Shear Load	99.62	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(189/2 + 13.2 + 14)\sin(120/2), 2 \cdot 103)$
 $= 210.712 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/(3\text{L}))]$
 $= -1514 \cdot 1200 [1 - (1 - 1200/5785 + (103^2 - 0^2)/(2 \cdot 1200 \cdot 5785)) / (1 + (4 \cdot 0)/(3 \cdot 5785))]$
 $= -375.5 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/L)$



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$$= 1514 \cdot 5785 / 4 \cdot ((1 + 2(103^2 - 0^2)) / (5785^2)) / (1 + (4 \cdot 0) / (3 \cdot 5785)) - 4 \cdot 1200 / 5785$$

$$= 374.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) - 374 / (\pi \cdot 103^2 \cdot 10.2)$$

$$= 1.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) + 374 / (\pi \cdot 103^2 \cdot 10.2)$$

$$= 23.01 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) - 376 / (0.11 \cdot \pi \cdot 103^2 \cdot 10.2)$$

$$= 114.78 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) + 376 / (0.19 \cdot \pi \cdot 103^2 \cdot 10.2)$$

$$= -44.82 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1514 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3))$$

$$= 886.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 886 / (103 \cdot 10.2)$$

$$= 9.74 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(103 \cdot 10.2)}$$

$$= 25.203 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(103 \cdot 10.2)}, 2 \cdot 1200)$$

$$= 325.41 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(118 / 118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 1514 \cdot 0.1 / (325(10.2 + 1 \cdot 14))$$

$$= -0.14 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -1514 / (4(10.2 + 1 \cdot 14)325) -$$

$$3 \cdot 0.053 \cdot 1514 / (2(10.2 + 1 \cdot 14)^2)$$

$$= -2.49 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3450.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

$$= 0.13139E-04 \cdot 3450 (260 - 21.1)$$

$$= 10.828 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	213.6	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(97.5 \cdot 10.2)} + 330)10.2^3)/12$$

$$= 3.333 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3.33 + 38.6 (173 - 5.09)^2$$

$$= 10870.216 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3158735/183$$

$$= 172.889 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1514$$

$$= 308.1685 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.2097 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 128.1609 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 535 - 97.5 \cdot \sin(120/2) / 1.05$$

$$= 454.389 \text{ mm}$$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9, 2026

Bending Moment, $M = F_h \cdot d = 140.0310 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 0.4035 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 142.4010 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(1514 + 95) \cdot 300) / (4 \cdot 600 \cdot 142)}$

$= 6.447 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$

$= 210.712 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 211 / (3 - 1)$

$= 105.356 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 105 \cdot 300 / 2$

$= 158.034 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 1609 / (600 \cdot 300)$

$= 0.894 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 158 \cdot 0.89$

$= 141.282 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 36.5 + 7.37$

$= 43.915 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 141 / 43.9$

$= 0.316 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 275 - 14$

$= 261.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 43.9$

$= 0.000 \text{ mm}$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2426/43.9}$$

$$= 74.330 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 535 - \cos(120/2) (94.5 + 13.2 + 14) - 14$$

$$= 460.173 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{214}}$$

$$= 135.934$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/74.3$$

$$= 6.191$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 99.6 / (2 \cdot 600) \cdot 105 \cdot 460/2$$

$$= 2.012 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($6.19 < 136$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.19)^2 / (2 \cdot 136^2)) 214 / (5/3 + 3 \cdot (6.19) / (8 \cdot 136) - (6.19^3) / (8 \cdot 136^3))$$

$$= 126.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.32/127 + (2.01 \cdot 138/24263022) / 142$$

$$= 0.003$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/51.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 535 - \cos(122/2) (14 + 0 + 0) - 0$$

$$= 399.345 \text{ mm}$$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{2428/51.3}$
 $= 68.809 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 399/68.8$
 $= 5.804$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 99.6 / (2 \cdot 600)$
 $= 0.083 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.083 \cdot 105 \cdot 399$
 $= 3.493 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 3.49/176610$
 $= 0.194 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($5.8 < 136$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (5.8)^2 / (2 \cdot 136^2)) 214 /$
 $(5/3 + 3 \cdot (5.8) / (8 \cdot 136) - (5.8)^3 / (8 \cdot 136^3))$
 $= 126.8 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 0.54/127 + (3.49 \cdot 138/2428) / 142$
 $= 0.006$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	137.60 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load Q0 (Weight)	Q0	986.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	15.4 Kg
Maximum Transverse Force	Ft	81.7 Kg
Maximum Longitudinal Force	F1	99.6 Kg
Saddle Bolted to Steel Foundation	No	No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 99.6 / (0.96 \cdot 4)$
 $= 2.5 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 81.7 / (0.96 \cdot 4)$
 $= 2.1 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $99.6 / (82.8 \cdot 4)$
= 0.0295 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $535 \cdot 81.7 + 0$
= 43.74 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= 43.7/987
= 44.33 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.029, 0]
= 0.0295 cm²

MDMT Calculations for: LH floating sad, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 13.2, tr = 4.24, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.42$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		134.60	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Saddle Force Q, Operating Case 1478.51 Kgf
Pressure used in Analysis P 2400.887 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	1.44	118.00
Long. Stress at Bottom of Midspan	22.75	118.00
Long. Stress at Top of Saddles	112.36	118.00
Long. Stress at Bottom of Saddles	-43.48	-176.76
Long. Stress at Bottom of Saddles	43.48	118.00
Tangential Shear in Shell	9.51	94.40
Circ. Stress at Horn of Saddle	2.43	147.50
Circ. Compressive Stress in Shell	0.14	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (163/2 + 0) • 535/211
= 622.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(16.8, 0) • 535/3450
= 2.6 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(99.6, 0) • 535/3450
= 15.4 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (99.6/2 + 0) • 535/211
= 379.4 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 856 + max(2.6, 623, 15.4, 379)
= 1478.5 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766 (0 + $\pi/4$ (0.42 • 1.2)²)
= 164.386 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1573.52	Kgf
Transverse Shear Load	81.75	Kgf
Longitudinal Shear Load	99.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(189/2 + 13.2 + 14)sin(120/2), 2•103)
= 210.712 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9, 2026

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -1479 \cdot 1200 \left[1 - \left(1 - \frac{1200}{5785} + \frac{(103^2 - 0^2)}{(2 \cdot 1200 \cdot 5785)} \right) / \left(1 + \frac{(4 \cdot 0)}{3 \cdot 5785} \right) \right]$$

$$= -366.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 1479 \cdot 5785 / 4 \left(\frac{(1 + 2(103^2 - 0^2)/(5785^2))}{(1 + (4 \cdot 0)/(3 \cdot 5785))} - \frac{4 \cdot 1200}{5785} \right)$$

$$= 365.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) - 365 / (\pi \cdot 103^2 \cdot 10.2)$$

$$= 1.44 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) + 365 / (\pi \cdot 103^2 \cdot 10.2)$$

$$= 22.75 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) - 367 / (0.11 \cdot \pi \cdot 103^2 \cdot 10.2)$$

$$= 112.36 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2401 \cdot 103 / (2 \cdot 10.2) + 367 / (0.19 \cdot \pi \cdot 103^2 \cdot 10.2)$$

$$= -43.48 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1479 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3))$$

$$= 865.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 865 / (103 \cdot 10.2)$$

$$= 9.51 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(103 \cdot 10.2)}$$

$$= 25.203 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(103 \cdot 10.2)}, 2 \cdot 1200)$$

$$= 325.41 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(118/118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:



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$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 1479 \cdot 0.1 / (325 (10.2 + 1 \cdot 14)) \\
 &= -0.14 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_r, r]$:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -1479 / (4(10.2 + 1 \cdot 14)325) - \\
 &\quad 3 \cdot 0.053 \cdot 1479 / (2(10.2 + 1 \cdot 14)^2) \\
 &= -2.43 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	213.6	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(97.5 \cdot 10.2)} + 330)10.2^3) / 12 \\
 &= 3.333 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 3.33 + 38.6 (173 - 5.09)^2 \\
 &= 10870.216 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 3158735 / 183 \\
 &= 172.889 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 1479 \\
 &= 300.9087 \text{ Kgf}
 \end{aligned}$$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Tension Stress, $St = (Fh/As) = 0.2048 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 128.1609 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 535 - 97.5 \cdot \sin(120/2)/1.05$
 $= 454.389 \text{ mm}$

Bending Moment, $M = Fh * d = 136.7321 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 0.3940 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 142.4010 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(1479 + 95)300) / (4 * 600 * 142)}$
 $= 6.375 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (94.5 + 13.2 + 14)$
 $= 210.712 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 211 / (3 - 1)$
 $= 105.356 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 105 * 300 / 2$
 $= 158.034 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 1574 / (600 * 300)$
 $= 0.874 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 158 * 0.87$
 $= 138.150 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 7.37$
 $= 43.915 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 138/43.9$
 $= 0.309 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

= 275 - 14
= 261.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/43.9
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 275/2
= 137.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{2426/43.9}$
= 74.330 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 535 - cos(120/2)(94.5 + 13.2 + 14) - 14
= 460.173 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/214}$
= 135.934

Slenderness ratio [KL/r]:

= KL/r
= 1 • 460/74.3
= 6.191

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 99.6 / (2 • 600) • 105 • 460/2
= 2.012 Kg-m.

Compressive Allowable, KL/r < Cc (6.19 < 136) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (6.19)^2 / (2 \cdot 136^2)) 214 /$
 $(5/3 + 3 \cdot (6.19) / (8 \cdot 136) - (6.19^3) / (8 \cdot 136^3))$
= 126.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
= 0.31/127 + (2.01 • 138/24263022)/142
= 0.003

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/51.3
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

= 275/2
= 137.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 535 - cos(122/2)(14 + 0 + 0) - 0
= 399.345 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{(2428/51.3)}$
= 68.809 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 • 399/68.8
= 5.804

Unit Force [Force,u]:
= FI / (2 • Baseplate Length)
= 99.6/(2 • 600)
= 0.083 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force • e • L
= 0.083 • 105 • 399
= 3.493 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 3.49/176610
= 0.194 N./mm²

Compressive Allowable, KL/r < Cc (5.8 < 136) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (5.8)^2 / (2 \cdot 136^2)) 214 / (5/3 + 3 \cdot (5.8) / (8 \cdot 136) - (5.8)^3 / (8 \cdot 136^3))$
= 126.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c • C1/I)/Sba
= 0.53/127 + (3.49 • 138/2428)/142
= 0.006

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	137.60 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load Q0 (Weight)	Q0	950.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	15.4 Kg
Maximum Transverse Force	Ft	81.7 Kg
Maximum Longitudinal Force	F1	99.6 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 296.0 Kg



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Saddle Calcs: Operating Case: Step: 20 9:39am Sep 9,2026

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(F_f; F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 296 / (0.96 \cdot 4)$$

$$= 7.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 81.7 / (0.96 \cdot 4)$$

$$= 2.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 99.6 / (82.8 \cdot 4)$$

$$= 0.0295 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 535 \cdot 81.7 + 0$$

$$= 43.74 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 43.7 / 951$$

$$= 46.00 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.029, 0]$$

$$= 0.0295 \text{ cm}^2$$

MDMT Calculations for: RH fixed saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 13.2, tr = 4.24, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.42, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

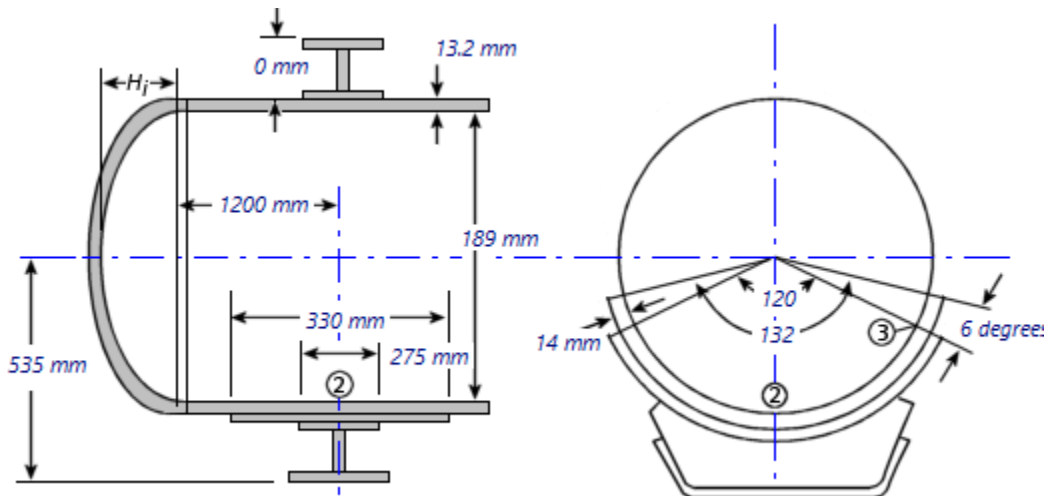
Step: 21 9:39am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		216.00	N./mm ²
Head Allowable Stress used in Calculation		216.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1050.38	Kgf
Pressure used in Analysis	P	3121.005	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	8.16	216.00
Long. Stress at Bottom of Midspan	23.29	216.00
Long. Stress at Top of Saddles	86.96	216.00
Long. Stress at Bottom of Saddles	-23.76	-216.00
Long. Stress at Bottom of Saddles	23.76	216.00
Tangential Shear in Shell	6.76	127.92
Circ. Stress at Horn of Saddle	1.72	323.99



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Saddle Calcs: Test Case:

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Circ. Compressive Stress in Shell	0.10	216.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (81.7/2 + 0) \cdot 535/211$$

$$= 311.3 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(8.38, 0) \cdot 535/3450$$

$$= 1.3 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 739 + \max(1.3, 311, 0, 0)$$

$$= 1050.4 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (0.42 \cdot 1.2)^2)$$

$$= 164.386 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1145.39	Kgf
Transverse Shear Load	40.87	Kgf
Longitudinal Shear Load	8.38	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(189/2 + 13.2 + 14) \sin(120/2), 2 \cdot 103)$$

$$= 210.712 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -1050 \cdot 1200 [1 - (1 - 1200/5785 + (103^2 - 0^2)/(2 \cdot 1200 \cdot 5785)) / (1 + (4 \cdot 0)/(3 \cdot 5785))]$$

$$= -260.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$$

$$= 1050 \cdot 5785/4 ((1 + 2(103^2 - 0^2)/(5785^2)) / (1 + (4 \cdot 0)/(3 \cdot 5785)) - 4 \cdot 1200/5785)$$

$$= 259.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 3121 \cdot 103/(2 \cdot 10.2) - 260/(\pi \cdot 103^2 \cdot 10.2)$$

$$= 8.16 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 3121 \cdot 103 / (2 \cdot 10.2) + 260 / (\pi \cdot 103^2 \cdot 10.2) \\
 &= 23.29 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3121 \cdot 103 / (2 \cdot 10.2) - 261 / (0.11 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= 86.96 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3121 \cdot 103 / (2 \cdot 10.2) + 261 / (0.19 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= -23.76 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 1050 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3)) \\
 &= 614.6 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K_2 \cdot T / (R_m \cdot t) \\
 &= 1.17 \cdot 615 / (103 \cdot 10.2) \\
 &= 6.76 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(103 \cdot 10.2)} \\
 &= 25.203 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(275 + 1.56 \cdot \sqrt{(103 \cdot 10.2)}, 2 \cdot 1200) \\
 &= 325.41 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r)) \\
 &= -0.76 \cdot 1050 \cdot 0.1 / (325(10.2 + 1 \cdot 14)) \\
 &= -0.10 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot t_r)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2) \\
 &= -1050 / (4(10.2 + 1 \cdot 14)325) - \\
 &\quad 3 \cdot 0.053 \cdot 1050 / (2(10.2 + 1 \cdot 14)^2) \\
 &= -1.72 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(97.5 \cdot 10.2)} + 330)10.2^3) / 12
 \end{aligned}$$



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$$= 3.333 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 3.33 + 38.6 (173 - 5.09)^2$$

$$= 10870.216 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3158735/183$$

$$= 172.889 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1050$$

$$= 213.7749 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1455 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 535 - 97.5 \cdot \sin(120/2) / 1.05$$

$$= 454.389 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 97.1388 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2799 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1050 + 95)300) / (4 \cdot 600 \cdot 173)}$$

$$= 4.930 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$$

$$= 210.712 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 211 / (3 - 1)$$

$$= 105.356 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 105 \cdot 300 / 2$$

$$= 158.034 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 1145 / (600 \cdot 300)$$

$$= 0.636 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 158 \cdot 0.64$$

$$= 100.562 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.5 + 7.37$$

$$= 43.915 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 101 / 43.9$$

$$= 0.225 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 43.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 275 / 2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (2426 / 43.9) }$$

$$= 74.330 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 535 - \cos(120 / 2) (94.5 + 13.2 + 14) - 14$$

$$= 460.173 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:



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$$= KL/r$$

$$= 1 \cdot 460/74.3$$

$$= 6.191$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 8.38 / (2 \cdot 600) \cdot 105 \cdot 460 / 2$$

$$= 0.169 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($6.19 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.19)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (6.19) / (8 \cdot 123) - (6.19^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.22/154 + (0.17 \cdot 138/24263022) / 173$$

$$= 0.002$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/51.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 535 - \cos(122/2) (14 + 0 + 0) - 0$$

$$= 399.345 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2428/51.3) }$$

$$= 68.809 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 399/68.8$$

$$= 5.804$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8.38 / (2 \cdot 600)$$

$$= 0.007 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.007 \cdot 105 \cdot 399$$

$$= 0.294 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:



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= Bending Moment / Section Modulus

= 0.29/176610

= 0.016 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.8 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (5.8)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (5.8) / (8 \cdot 123) - (5.8^3) / (8 \cdot 123^3))$

= 154.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$

= $0.38 / 154 + (0.29 \cdot 138 / 2428) / 173$

= 0.003

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load QO (Weight)	QO	834.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1.3 Kg
Maximum Transverse Force	Ft	40.9 Kg
Maximum Longitudinal Force	Fl	8.4 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $8.38 / (0.96 \cdot 4)$

= 0.2 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $40.9 / (0.96 \cdot 4)$

= 1.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $8.38 / (29 \cdot 4)$

= 0.0071 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $535 \cdot 40.9 + 0$

= 21.87 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= 21.9/834

= 26.22 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:



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= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.0071, 0]

= 0.0071 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		216.00	N./mm ²
Head Allowable Stress used in Calculation		216.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1021.46	Kgf
Pressure used in Analysis	P	3121.005	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	8.36	216.00
Long. Stress at Bottom of Midspan	23.08	216.00
Long. Stress at Top of Saddles	84.99	216.00
Long. Stress at Bottom of Saddles	-22.67	-216.00
Long. Stress at Bottom of Saddles	22.67	216.00
Tangential Shear in Shell	6.57	127.92
Circ. Stress at Horn of Saddle	1.68	323.99
Circ. Compressive Stress in Shell	0.10	216.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (81.7/2 + 0) • 535/211

= 311.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(8.38, 0) • 535/3450

= 1.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 710 + \max(1.3, 311, 0, 0) \\
 &= 1021.5 \text{ Kg}
 \end{aligned}$$

Longitudinal Wind Force [F_L]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766 (0 + \pi/4 (0.42 \cdot 1.2)^2) \\
 &= 164.386 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1116.47	Kgf
Transverse Shear Load	40.87	Kgf
Longitudinal Shear Load	8.38	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(189/2 + 13.2 + 14) \sin(120/2), 2 \cdot 103) \\
 &= 210.712 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -1021 \cdot 1200 [1 - (1 - 1200/5785 + (103^2 - 0^2)/(2 \cdot 1200 \cdot 5785))/(1 + (4 \cdot 0)/(3 \cdot 5785))] \\
 &= -253.3 \text{ Kg-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 1021 \cdot 5785/4 ((1 + 2(103^2 - 0^2)/(5785^2))/(1 + (4 \cdot 0)/(3 \cdot 5785)) - 4 \cdot 1200/5785) \\
 &= 252.5 \text{ Kg-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 3121 \cdot 103/(2 \cdot 10.2) - 252/(\pi \cdot 103^2 \cdot 10.2) \\
 &= 8.36 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 3121 \cdot 103/(2 \cdot 10.2) + 252/(\pi \cdot 103^2 \cdot 10.2) \\
 &= 23.08 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 3121 \cdot 103/(2 \cdot 10.2) - 253/(0.11 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= 84.99 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3121 \cdot 103/(2 \cdot 10.2) + 253/(0.19 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= -22.67 \text{ N/mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$



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$$= 1021 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0/3))$$

$$= 597.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 598 / (103 \cdot 10.2)$$

$$= 6.57 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ R_m \cdot t }$$

$$= 0.78 \cdot \sqrt{ 103 \cdot 10.2 }$$

$$= 25.203 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{ 103 \cdot 10.2 }, 2 \cdot 1200)$$

$$= 325.41 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot tr))$$

$$= -0.76 \cdot 1021 \cdot 0.1 / (325 (10.2 + 1 \cdot 14))$$

$$= -0.10 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -1021 / (4(10.2 + 1 \cdot 14) 325) -$$

$$3 \cdot 0.053 \cdot 1021 / (2(10.2 + 1 \cdot 14)^2)$$

$$= -1.68 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	440.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 97.5 \cdot 10.2 } + 330) 10.2^3) / 12$$

$$= 3.333 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 3.33 + 38.6 (173 - 5.09)^2$$

$$= 10870.216 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 3158735/183$$

$$= 172.889 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1021$$

$$= 207.8889 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1415 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 535 - 97.5 \cdot \sin(120/2) / 1.05$$

$$= 454.389 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 94.4642 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2722 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1021 + 95) \cdot 300) / (4 \cdot 600 \cdot 173)}$$

$$= 4.867 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$$

$$= 210.712 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 211 / (3 - 1)$$

$$= 105.356 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 105 \cdot 300 / 2$$

$$= 158.034 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1116 / (600 \cdot 300)$$

$$= 0.620 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 158 \cdot 0.62$$

$$= 98.022 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 36.5 + 7.37

= 43.915 cm²

Compressive Stress [Sc]:

= P/Ar

= 98/43.9

= 0.219 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 275 - 14

= 261.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/43.9

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 275/2

= 137.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2426/43.9}$

= 74.330 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 535 - cos(120/2)(94.5 + 13.2 + 14) - 14

= 460.173 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 460/74.3

= 6.191

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 8.38 / (2 • 600) • 105 • 460/2

= 0.169 Kg-m.

Compressive Allowable, KL/r < Cc (6.19 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (6.19)² / (2 • 123²)) 260 /

(5/3 + 3 • (6.19) / (8 • 123) - (6.19³) / (8 • 123³))

= 154.1 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.22/154 + (0.17 \cdot 138/24263022) / 173$$

$$= 0.001$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/51.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 535 - \cos(122/2) (14 + 0 + 0) - 0$$

$$= 399.345 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2428/51.3) }$$

$$= 68.809 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 399/68.8$$

$$= 5.804$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8.38 / (2 \cdot 600)$$

$$= 0.007 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.007 \cdot 105 \cdot 399$$

$$= 0.294 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0.29/176610$$

$$= 0.016 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($5.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (5.8) / (8 \cdot 123) - (5.8^3) / (8 \cdot 123^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 0.38/154 + (0.29 \cdot 138/2428) / 173$$

$$= 0.003$$



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Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load Q0 (Weight)	Q0	805.1 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1.3 Kgf
Maximum Transverse Force	Ft	40.9 Kgf
Maximum Longitudinal Force	Fl	8.4 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $8.38 / (0.96 \cdot 4)$

= 0.2 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $40.9 / (0.96 \cdot 4)$

= 1.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $8.38 / (29 \cdot 4)$

= 0.0071 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $535 \cdot 40.9 + 0$

= 21.87 Kg-m.

Eccentricity (e):

= $Rmom / Q0$

= $21.9 / 805$

= 27.16 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0, 0.0071, 0]$

= 0.0071 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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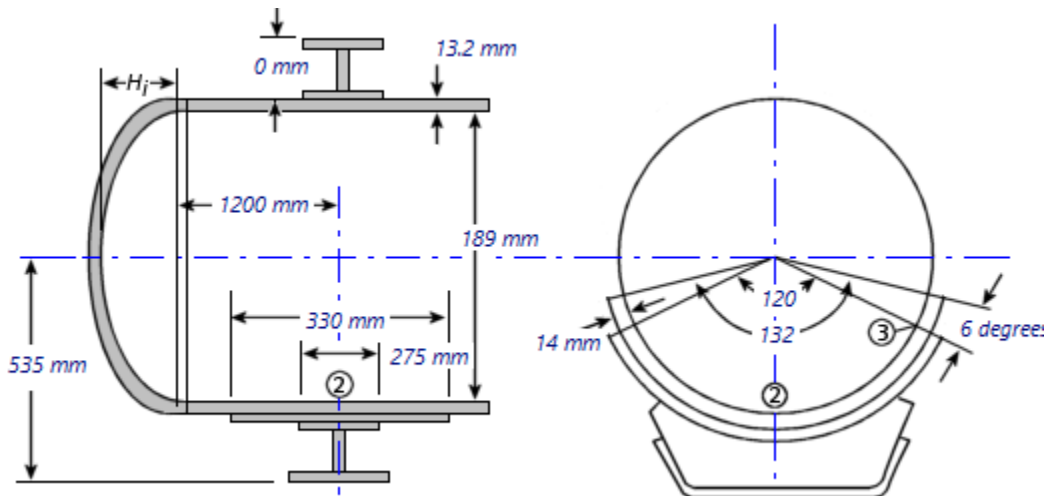
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		133.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	1514.18	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-11.43	-176.76
Long. Stress at Top of Midspan	11.43	118.00
Long. Stress at Bottom of Midspan	10.39	118.00
Long. Stress at Top of Saddles	102.16	118.00
Long. Stress at Bottom of Saddles	-57.43	-176.76
Long. Stress at Bottom of Saddles	57.43	118.00
Tangential Shear in Shell	9.74	94.40



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Circ. Stress at Horn of Saddle	2.49	147.50
Circ. Compressive Stress in Shell	0.14	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (163/2 + 0) \cdot 535/211$$

$$= 622.7 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind; Sum of X Forces}) \cdot B / Ls$$

$$= \max(16.8, 0) \cdot 535/3450$$

$$= 2.6 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / Ls$$

$$= \max(99.6, 0) \cdot 535/3450$$

$$= 15.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (99.6/2 + 0) \cdot 535/211$$

$$= 379.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 892 + \max(2.6, 623, 15.4, 379)$$

$$= 1514.2 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(0.42 \cdot 1.2)^2)$$

$$= 164.386 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1609.19	Kgf
Transverse Shear Load	81.75	Kgf
Longitudinal Shear Load	99.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(189/2 + 13.2 + 14)\sin(120/2), 2 \cdot 103)$$

$$= 210.712 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -1514 \cdot 1200 [1 - (1 - 1200/5785 + (103^2 - 0^2)/(2 \cdot 1200 \cdot 5785))/(1 + (4 \cdot 0)/(3 \cdot 5785))]$$

$$= -375.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 1514 \cdot 5785 / 4 \cdot ((1 + 2(103^2 - 0^2) / (5785^2)) / (1 + (4 \cdot 0) / (3 \cdot 5785)) - 4 \cdot 1200 / 5785) \\
 &= 374.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 103 / (2 \cdot 10.2) - 374 / (\pi \cdot 103^2 \cdot 10.2) \\
 &= -11.43 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 103 / (2 \cdot 10.2) + 374 / (\pi \cdot 103^2 \cdot 10.2) \\
 &= 10.39 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 103 / (2 \cdot 10.2) - 376 / (0.11 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= 102.16 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 103 / (2 \cdot 10.2) + 376 / (0.19 \cdot \pi \cdot 103^2 \cdot 10.2) \\
 &= -57.43 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 1514 \cdot (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3)) \\
 &= 886.0 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 886 / (103 \cdot 10.2) \\
 &= 9.74 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(103 \cdot 10.2)} \\
 &= 25.203 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(275 + 1.56 \cdot \sqrt{(103 \cdot 10.2)}, 2 \cdot 1200) \\
 &= 325.41 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(118 / 118, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 1514 \cdot 0.1 / (325(10.2 + 1 \cdot 14)) \\
 &= -0.14 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -1514 / (4(10.2 + 1 \cdot 14)325) - \\
 &\quad 3 \cdot 0.053 \cdot 1514 / (2(10.2 + 1 \cdot 14)^2) \\
 &= -2.49 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm



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Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	213.6	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(97.5 \cdot 10.2)} + 330)10.2^3)/12$$

$$= 3.333 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3.33 + 38.6(173 - 5.09)^2$$

$$= 10870.216 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3158735/183$$

$$= 172.889 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1514$$

$$= 308.1685 \text{ Kgf}$$

Tension Stress, St = (Fh/As)	=	0.2097	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	128.1609	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 535 - 97.5 \cdot \sin(120/2) / 1.05$$

$$= 454.389 \text{ mm}$$

Bending Moment, M = Fh * d	=	140.0310	Kg-m.
----------------------------	---	----------	-------

Bending Stress, Sb = (M * C1 / I)	=	0.4035	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	142.4010	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$



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$$= \sqrt{3(1514 + 95)300 / (4 \cdot 600 \cdot 142)}$$

$$= 6.447 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$$

$$= 210.712 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 211 / (3 - 1)$$

$$= 105.356 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 105 \cdot 300 / 2$$

$$= 158.034 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1609 / (600 \cdot 300)$$

$$= 0.894 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 158 \cdot 0.89$$

$$= 141.282 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 7.37$$

$$= 43.915 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 141 / 43.9$$

$$= 0.316 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 43.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275 / 2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



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$$= \sqrt{2426/43.9}$$

$$= 74.330 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 535 - \cos(120/2) (94.5 + 13.2 + 14) - 14$$

$$= 460.173 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943008 / 214}$$

$$= 135.934$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460 / 74.3$$

$$= 6.191$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 99.6 / (2 \cdot 600) \cdot 105 \cdot 460 / 2$$

$$= 2.012 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($6.19 < 136$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.19)^2 / (2 \cdot 136^2)) 214 / (5/3 + 3 \cdot (6.19) / (8 \cdot 136) - (6.19)^3 / (8 \cdot 136^3))$$

$$= 126.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.32 / 127 + (2.01 \cdot 138 / 24263022) / 142$$

$$= 0.003$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 51.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275 / 2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 535 - \cos(122/2) (14 + 0 + 0) - 0$$

$$= 399.345 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2428 / 51.3}$$

$$= 68.809 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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$$= 1 \cdot 399/68.8$$

$$= 5.804$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 99.6 / (2 \cdot 600)$$

$$= 0.083 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.083 \cdot 105 \cdot 399$$

$$= 3.493 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.49/176610$$

$$= 0.194 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.8 < 136$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.8)^2 / (2 \cdot 136^2)) 214 / (5/3 + 3 \cdot (5.8) / (8 \cdot 136) - (5.8^3) / (8 \cdot 136^3))$$

$$= 126.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 0.54/127 + (3.49 \cdot 138/2428) / 142$$

$$= 0.006$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	102.57	mm
Shell Thickness used in this Case	t	13.179	mm
Stiffened Vessel Length per 4.15.6	L	5785.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		134.60	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		1478.51	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-11.17	-176.76
Long. Stress at Top	of Midspan	11.17	118.00
Long. Stress at Bottom	of Midspan	10.13	118.00



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Long. Stress at Top of Saddles	99.74	118.00
Long. Stress at Bottom of Saddles	-56.09	-176.76
Long. Stress at Bottom of Saddles	56.09	118.00
Tangential Shear in Shell	9.51	94.40
Circ. Stress at Horn of Saddle	2.43	147.50
Circ. Compressive Stress in Shell	0.14	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (163/2 + 0) \cdot 535/211$$

$$= 622.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(16.8, 0) \cdot 535/3450$$

$$= 2.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(99.6, 0) \cdot 535/3450$$

$$= 15.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (99.6/2 + 0) \cdot 535/211$$

$$= 379.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 856 + \max(2.6, 623, 15.4, 379)$$

$$= 1478.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (0.42 \cdot 1.2)^2)$$

$$= 164.386 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1573.52	Kgf
Transverse Shear Load	81.75	Kgf
Longitudinal Shear Load	99.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(189/2 + 13.2 + 14) \sin(120/2), 2 \cdot 103)$$

$$= 210.712 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$



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$$= -1479 \cdot 1200 [1 - (1 - 1200/5785 + (103^2 - 0^2) / (2 \cdot 1200 \cdot 5785)) / (1 + (4 \cdot 0) / (3 \cdot 5785))] \\ = -366.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 1479 \cdot 5785 / 4 \cdot ((1 + 2(103^2 - 0^2) / (5785^2)) / (1 + (4 \cdot 0) / (3 \cdot 5785)) - 4 \cdot 1200 / 5785) \\ = 365.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ = -103 \cdot 103 / (2 \cdot 10.2) - 365 / (\pi \cdot 103^2 \cdot 10.2) \\ = -11.17 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 103 / (2 \cdot 10.2) + 365 / (\pi \cdot 103^2 \cdot 10.2) \\ = 10.13 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ = -103 \cdot 103 / (2 \cdot 10.2) - 367 / (0.11 \cdot \pi \cdot 103^2 \cdot 10.2) \\ = 99.74 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 103 / (2 \cdot 10.2) + 367 / (0.19 \cdot \pi \cdot 103^2 \cdot 10.2) \\ = -56.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 1479 (5785 - 2 \cdot 1200) / (5785 + (4 \cdot 0 / 3)) \\ = 865.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.17 \cdot 865 / (103 \cdot 10.2) \\ = 9.51 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(103 \cdot 10.2)} \\ = 25.203 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(275 + 1.56 \cdot \sqrt{(103 \cdot 10.2)}, 2 \cdot 1200) \\ = 325.41 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(118/118, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.76 \cdot 1479 \cdot 0.1 / (325(10.2 + 1 \cdot 14)) \\ = -0.14 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -1479 / (4(10.2 + 1 \cdot 14)325) - 3 \cdot 0.053 \cdot 1479 / (2(10.2 + 1 \cdot 14)^2) \\ = -2.43 \text{ N./mm}^2$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	600.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	213.6	N./mm ²
Height of Web at Center	Hw,c	440.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(97.5 \cdot 10.2)} + 330)10.2^3)/12$$

$$= 3.333 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3.33 + 38.6(173 - 5.09)^2$$

$$= 10870.216 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	379.1	10.2	5.1	38.6	19643.4	0.109E+05
Wearplate	330.0	14.0	17.2	46.2	79368.9	0.112E+05
Web	14.0	399.3	223.9	55.9	1251519.6	0.888E+04
BasePlate	300.0	14.0	430.5	42.0	1808203.6	0.279E+05
Totals	...	...	...	182.7	3158735.5	0.588E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3158735/183$$

$$= 172.889 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1479$$

$$= 300.9087 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.2048 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 128.1609 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 535 - 97.5 \cdot \sin(120/2) / 1.05$$

$$= 454.389 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 136.7321 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M \cdot C_1 / I) = 0.3940 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 142.4010 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(1479 + 95) \cdot 300}{4 \cdot 600 \cdot 142}}$$

$$= 6.375 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (94.5 + 13.2 + 14)$$

$$= 210.712 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 211 / (3 - 1)$$

$$= 105.356 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 105 \cdot 300 / 2$$

$$= 158.034 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1574 / (600 \cdot 300)$$

$$= 0.874 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 158 \cdot 0.87$$

$$= 138.150 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 7.37$$

$$= 43.915 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 138 / 43.9$$

$$= 0.309 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 43.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:



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= Saddle Width / 2

= 275/2

= 137.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2426/43.9}$

= 74.330 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 535 - $\cos(120/2)$ (94.5 + 13.2 + 14) - 14

= 460.173 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/214}$

= 135.934

Slenderness ratio [KL/r]:

= KL/r

= 1 • 460/74.3

= 6.191

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= 99.6 / (2 • 600) • 105 • 460/2

= 2.012 Kg-m.

Compressive Allowable, $KL/r < Cc$ (6.19 < 136) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.19)^2 / (2 \cdot 136^2)) 214 /$

$(5/3 + 3 \cdot (6.19) / (8 \cdot 136) - (6.19)^3 / (8 \cdot 136^3))$

= 126.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$

= 0.31/127 + (2.01 • 138/24263022) / 142

= 0.003

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	105.4	14.0	0.0	14.7	0.0	2.41
Totals	...	...	...	51.3	...	0.243E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/51.3

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 275/2

= 137.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 535 - $\cos(122/2)$ (14 + 0 + 0) - 0

= 399.345 mm

Radius of Gyration [r]:



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Saddle Calcs: Ext. Pressure:

Step: 22

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$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{2428/51.3} \\
 &= 68.809 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 399/68.8 \\
 &= 5.804
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 99.6 / (2 \cdot 600) \\
 &= 0.083 \text{ Kg/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.083 \cdot 105 \cdot 399 \\
 &= 3.493 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 3.49/176610 \\
 &= 0.194 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($5.8 < 136$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (5.8)^2 / (2 \cdot 136^2)) 214 / \\
 &\quad (5/3 + 3 \cdot (5.8) / (8 \cdot 136) - (5.8^3) / (8 \cdot 136^3)) \\
 &= 126.8 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 0.53/127 + (3.49 \cdot 138/2428) / 142 \\
 &= 0.006
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

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Input, Nozzle Desc: N1-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	2400.000	KPa.
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	270	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	188.95	mm
Design Length of Section	L	366.5459	mm
Shell Finished (Minimum) Thickness	t	13.1794	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		200.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		89.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.20	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	129.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		20.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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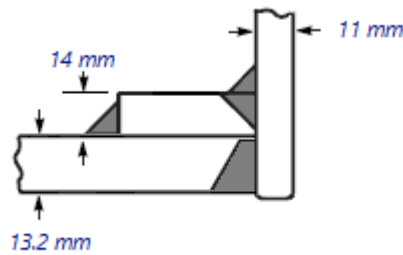
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Nozzle Calcs.: N1-Tube Inlet

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	89.000 mm.
Actual Thickness Used in Calculation	11.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2400 \cdot 97.5 / (118 \cdot 1 - 0.6 \cdot 2400)$
 $= 2.0072 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2400 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 2400)$
 $= 0.8978 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4125 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.0000 mm
Parallel to Vessel Wall, opening length	d	73.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.4486 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1-Tube Inlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8978 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0072 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.0072 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(5.01, 4.5)]$
 $= 5.0072 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9, 5.01)$
 $= 5.0072 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK



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Nozzle Calcs.: N1-Tube Inlet

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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	49.2,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	245.8 N./mm ²	Passed
Occasional :	4.9,	Allowable :	156.9 N./mm ²	Passed
Shear :	30.0,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 13.2, tr = 2.01, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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 Nozzle Calcs.: N1-Tube Inlet Nozl: 9 9:39am Sep 9,2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2400.00/10200.00 = 0.235

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0622 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	8.4840 = 0.7 * Wo mm
Pad Weld	6.0311 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11594 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.1366 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.3161 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	188.951	mm
Shell Thickness	13.1794	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	89.000	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	129.000	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2400.00	KPa.
Radial Load	P	-734.25	Kgf
Circumferential Shear	Vc	530.29	Kgf
Longitudinal Shear	Vl	530.29	Kgf
Circumferential Moment	Mc	130.48	Kg-m.



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Nozzle Calcs.: N1-Tube Inlet

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Longitudinal Moment	Ml	130.48	Kg-m.
Torsional Moment	Mt	150.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.337) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (8.091) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{109.565(13.179 - 3.0)}$$

$$= 199.208 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	219.130 mm
Nozzle Outside Diameter	(d) =	89.000 mm
Vessel Thickness used	(T) =	24.179 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	3.022
d / t	=	11.125

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.223$$

Nr/P	=	0.148
Mr/P	=	0.117
M0/P	=	0.131
N0/P	=	0.194
MrD/Mc	=	0.176
NrDL/Mc	=	0.175
M0d/Mc	=	0.199
N0DL/Mc	=	0.052
MrD/Ml	=	0.143
NrDL/Ml	=	0.147
M0D/Ml	=	0.173
N0DL/Ml	=	0.173

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	8	11	8	11
Outplane Stress Summary	-9	24	50	-17

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8



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Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary	-10	15	39	-20
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	15	28	51	22
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4

Outplane Stress Summary	-14	25	47	-31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-25	25	25	-25
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	8	11	8	11

Inplane Stress Summary	-11	25	49	-17
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4



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Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	16	28	54	33

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	9	12	18	21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-33	-33	33	33
Axial Bending (ML)	-93	93	93	-93
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-89	40	166	-79

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	19	19	10	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	106	50	166	103

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



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Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	12	15	15	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	-33	-33	33	33
Axial Bending (Mc)	-119	119	119	-119
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-114	66	191	-105

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14
Shear Stress Summary	10	10	19	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	129	68	193	129

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{102.565(13.179 - 3.0)}$$

$$= 195.629 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	205.130 mm
Nozzle Outside Diameter	(d) =	129.000 mm
Vessel Thickness used	(T) =	10.179 mm
Nozzle Thickness used	(t) =	28.000 mm
T / t	=	0.364
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.823$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.110
M0/P	=	0.034
N0/P	=	0.077
MrD/Mc	=	0.268
NrDL/Mc	=	0.096
M0d/Mc	=	0.081
N0DL/Mc	=	0.085



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MrD/M1 = 0.092
NrDL/M1 = 0.058
MØD/M1 = 0.026
NØDL/M1 = 0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	21	24	21	24
Outplane Stress Summary	18	21	64	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	2	16	119	-78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	22	27	119	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	10	10	10	10



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Outplane Stress Summary	-23	40	85	-36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	24	21	24

Inplane Stress Summary	-91	127	234	-162
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	8	8
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	91	127	235	162
------------------------	----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	11	8	11	8	11	8	11
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	5	5
Circ. Q (SUS)	-16	16	34	-34	-17	17	33	-33
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-2	-2	4	4	1	1	3	3
Long. Q (SUS)	-13	13	29	-29	-20	20	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	8.0	11.0	8.0	11.0	8.0	11.0	8.0	11.0
Pm+Pl (SUS)	6.2	9.1	14.2	17.1	6.0	8.3	13.2	16.1
Pm+Pl+Q (Total)	15.5	27.1	48.8	22.6	16.9	28.0	50.5	32.6

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	17.11	177.00	Passed
Pm+Pl+Q (TOTAL)	50.52	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	-3	-3	13	13	-5	-5	13	13
Circ. Q (SUS)	0	0	28	-28	-108	108	198	-198
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	9	9	-3	-3	13	13
Long. Q (SUS)	-8	8	98	-98	-32	32	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+Pl (SUS)	18.4	21.3	34.3	37.2	16.4	19.3	34.4	37.3
Pm+Pl+Q (Total)	22.0	26.3	117.2	88.2	92.1	127.1	232.3	161.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+Pl (SUS)	37.28	177.00	Passed
Pm+Pl+Q (TOTAL)	232.33	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	14	11	14	11	14	11	14
Circ. Pl (SUS)	-2	-2	6	6	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-30	-30	36	36	-30	-30	36	36
Long. Q (SUS)	-65	65	121	-121	-91	91	147	-147
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4



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Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	11.0	14.0	11.0	14.0	11.0	14.0	11.0	14.0
Pm+Pl (SUS)	34.9	37.9	41.6	41.7	37.9	40.8	41.6	41.6
Pm+Pl+Q (Total)	105.3	48.8	162.7	102.0	129.6	67.9	189.8	128.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	41.74	177.00	Passed
Pm+Pl+Q (TOTAL)	189.84	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N3-Shell inlet

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Input, Nozzle Desc: N3-Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	2400.000	KPa.
Temperature for Internal Pressure	Temp	260	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	260	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	188.95	mm
Design Length of Section	L	5785.0005	mm
Shell Finished (Minimum) Thickness	t	13.1794	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1105.97	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		89.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.60	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	129.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		20.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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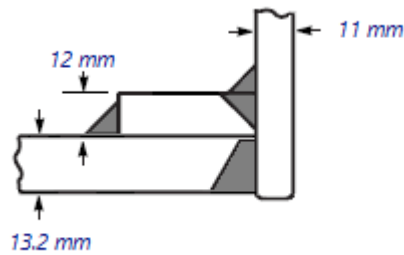
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Nozzle Calcs.: N3-Shell inlet

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Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	89.000 mm.
Actual Thickness Used in Calculation	11.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2400 \cdot 97.5 / (118 \cdot 1 - 0.6 \cdot 2400)$
 $= 2.0072 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2400 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 2400)$
 $= 0.8978 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4111 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.0000 mm
Parallel to Vessel Wall, opening length	d	73.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.4486 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3-Shell inlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8978 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0072 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.0072 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(5.01, 4.5)]$
 $= 5.0072 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9, 5.01)$
 $= 5.0072 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK



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Nozzle Calcs.: N3-Shell inlet

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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	49.2,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	245.8 N./mm ²	Passed
Occasional :	4.9,	Allowable :	156.9 N./mm ²	Passed
Shear :	30.0,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Nozzle Calcs.: N3-Shell inlet Nozl: 10 9:39am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2400.00/10200.00 = 0.235

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11594 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.1366 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.3161 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	188.951	mm
Shell Thickness	13.1794	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	89.000	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	129.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2400.00	KPa.
Radial Load	P	-734.25	Kgf
Circumferential Shear	Vc	530.29	Kgf
Longitudinal Shear	Vl	530.29	Kgf



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Nozzle Calcs.: N3-Shell inlet

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Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	130.48	Kg-m.
Torsional Moment	Mt	150.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (16.337) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (8.091) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{108.565(13.179 - 3.0)}$$

$$= 198.703 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	217.130 mm
Nozzle Outside Diameter	(d) =	89.000 mm
Vessel Thickness used	(T) =	22.179 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	2.772
d / t	=	11.125

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.282$$

Nr/P	=	0.144
Mr/P	=	0.118
M0/P	=	0.115
N0/P	=	0.184
MrD/Mc	=	0.184
NrDL/Mc	=	0.178
M0d/Mc	=	0.187
N0DL/Mc	=	0.055
MrD/Ml	=	0.147
NrDL/Ml	=	0.146
M0D/Ml	=	0.158
N0DL/Ml	=	0.173

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	9	11	9	11
Outplane Stress Summary	-10	27	54	-18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2



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Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-12	18	47	-25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1	
Outplane Shear (Vl)	0	0	0	0	
Outplane Shear (Mt)	5	5	5	5	
Shear Stress Summary	7	7	3	3	

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	18	30	56	26	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2	
Outplane Bending (P)	10	-10	10	-10	
Outplane Membrane (Mc)	-1	-1	1	1	
Outplane Bending (Mc)	-32	32	32	-32	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	4	4	4	4	
Outplane Stress Summary	-16	28	51	-33	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2	
Inplane Bending (P)	10	-10	10	-10	
Inplane Membrane (Mc)	-5	-5	5	5	
Inplane Bending (Mc)	-32	32	32	-32	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	9	11	9	11	
Inplane Stress Summary	-15	30	59	-23	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	-1	-1	1	1	
Torsional Shear (Mt)	5	5	5	5	



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Shear Stress Summary	3	3	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	33	63	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	9	12	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-33	-33	33	33
Axial Bending (ML)	-100	100	100	-100
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-93	45	174	-87

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	19	19	10	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	54	174	111

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0



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Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14

Hoop Stress Summary	12	15	16	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-33	-33	33	33
Axial Bending (Mc)	-128	128	128	-128
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-122	73	202	-116
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14

Shear Stress Summary	10	10	19	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	136	75	204	140
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{102.565(13.179 - 3.0)}$$

$$= 195.629 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	205.130 mm
Nozzle Outside Diameter	(d) =	129.000 mm
Vessel Thickness used	(T) =	10.179 mm
Nozzle Thickness used	(t) =	28.000 mm
T / t	=	0.364
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.823$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.110
M0/P	=	0.034
N0/P	=	0.077
MrD/Mc	=	0.268
NrDL/Mc	=	0.096
M0d/Mc	=	0.081



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NØDL/Mc = 0.085
MrD/Ml = 0.092
NrDL/Ml = 0.058
MØD/Ml = 0.026
NØDL/Ml = 0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	21	24	21	24
Outplane Stress Summary	18	21	64	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	2	16	119	-78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	22	27	119	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Normal Pressure Stress	10	10	10	10
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Outplane Stress Summary	-23	40	85	-36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	24	21	24

Inplane Stress Summary	-91	127	234	-162
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	91	127	235	162
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	11	9	11	9	11	9	11
Circ. Pl (SUS)	-3	-3	7	7	-3	-3	7	7
Circ. Q (SUS)	-17	17	37	-37	-22	22	42	-42
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-2	-2	6	6	1	1	3	3
Long. Q (SUS)	-15	15	35	-35	-22	22	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	9.0	11.0	9.0	11.0	9.0	11.0	9.0	11.0
Pm+Pl (SUS)	6.2	8.2	16.2	18.1	6.6	8.3	16.1	18.1
Pm+Pl+Q (Total)	18.1	28.2	54.7	27.0	20.5	32.8	61.0	37.6

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	18.12	177.00	Passed
Pm+Pl+Q (TOTAL)	61.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	-3	-3	13	13	-5	-5	13	13
Circ. Q (SUS)	0	0	28	-28	-108	108	198	-198
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	9	9	-3	-3	13	13
Long. Q (SUS)	-8	8	98	-98	-32	32	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+Pl (SUS)	18.4	21.3	34.3	37.2	16.4	19.3	34.4	37.3
Pm+Pl+Q (Total)	22.0	26.3	117.2	88.2	92.1	127.1	232.3	161.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+Pl (SUS)	37.28	177.00	Passed
Pm+Pl+Q (TOTAL)	232.33	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	14	11	14	11	14	11	14
Circ. Pl (SUS)	-3	-3	7	7	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-30	-30	36	36	-30	-30	36	36
Long. Q (SUS)	-70	70	130	-130	-98	98	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Shear P1 (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	11.0	14.0	11.0	14.0	11.0	14.0	11.0	14.0
Pm+P1 (SUS)	34.0	36.9	41.7	41.8	37.9	40.8	41.6	41.6
Pm+P1+Q (Total)	109.1	52.8	171.7	111.8	136.5	74.7	200.7	138.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+P1 (SUS)	41.77	177.00	Passed
Pm+P1+Q (TOTAL)	200.74	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Shell outlet

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Input, Nozzle Desc: N4-Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	2401.813	KPa.
Temperature for Internal Pressure	Temp	260	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	260	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	188.95	mm
Design Length of Section	L	5785.0005	mm
Shell Finished (Minimum) Thickness	t	13.1794	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5755.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		89.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.60	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	129.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		20.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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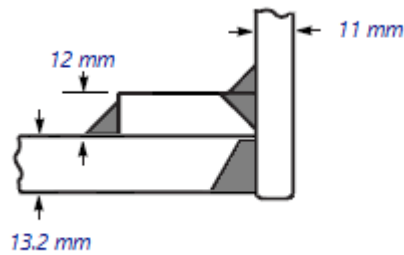
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Nozzle Calcs.: N4-Shell outlet

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Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	89.000 mm.
Actual Thickness Used in Calculation	11.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2402 \cdot 97.5 / (118 \cdot 1 - 0.6 \cdot 2402)$
 $= 2.0087 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2402 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 2402)$
 $= 0.8985 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4111 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.0000 mm
Parallel to Vessel Wall, opening length	d	73.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.4486 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-Shell outlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8985 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0087 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.0087 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(5.01, 4.5)]$
 $= 5.0087 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9, 5.01)$
 $= 5.0087 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK



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Nozzle Calcs.: N4-Shell outlet

Nozl: 11

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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	49.2,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	245.8 N./mm ²	Passed
Occasional	:	4.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	30.0,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Nozzle Calcs.: N4-Shell outlet

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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2401.81/10200.00 = 0.235$

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11594 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.1366 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.3161 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	188.951	mm
Shell Thickness	13.1794	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	89.000	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	129.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2401.81	KPa.
Radial Load	P	-734.25	Kgf
Circumferential Shear	Vc	530.29	Kgf
Longitudinal Shear	Vl	530.29	Kgf



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Nozzle Calcs.: N4-Shell outlet

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Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	130.48	Kg-m.
Torsional Moment	Mt	150.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.337) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (8.091) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{108.565(13.179 - 3.0)}$$

$$= 198.703 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	217.130 mm
Nozzle Outside Diameter	(d) =	89.000 mm
Vessel Thickness used	(T) =	22.179 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	2.772
d / t	=	11.125

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.282$$

Nr/P	=	0.144
Mr/P	=	0.118
M0/P	=	0.115
N0/P	=	0.184
MrD/Mc	=	0.184
NrDL/Mc	=	0.178
M0d/Mc	=	0.187
N0DL/Mc	=	0.055
MrD/Ml	=	0.147
NrDL/Ml	=	0.146
M0D/Ml	=	0.158
N0DL/Ml	=	0.173

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-5	-5	5	5
Outplane Bending (Ml)	-27	27	27	-27
Normal Pressure Stress	9	11	9	11
Outplane Stress Summary	-10	27	54	-18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2



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Nozzle Calcs.: N4-Shell outlet

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Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-12	18	47	-25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1	
Outplane Shear (Vl)	0	0	0	0	
Outplane Shear (Mt)	5	5	5	5	
Shear Stress Summary	7	7	3	3	

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	18	30	56	26	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2	
Outplane Bending (P)	10	-10	10	-10	
Outplane Membrane (Mc)	-1	-1	1	1	
Outplane Bending (Mc)	-32	32	32	-32	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	4	4	4	4	
Outplane Stress Summary	-16	28	51	-33	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2	
Inplane Bending (P)	10	-10	10	-10	
Inplane Membrane (Mc)	-5	-5	5	5	
Inplane Bending (Mc)	-32	32	32	-32	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	9	11	9	11	
Inplane Stress Summary	-15	30	59	-23	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	-1	-1	1	1	
Torsional Shear (Mt)	5	5	5	5	



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Nozzle Calcs.: N4-Shell outlet

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Shear Stress Summary	3	3	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	33	63	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	9	12	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-33	-33	33	33
Axial Bending (ML)	-100	100	100	-100
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-93	45	174	-87

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	19	19	10	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	54	174	111

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet Nozl: 11 9:39am Sep 9,2026

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14

Hoop Stress Summary	12	15	16	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-33	-33	33	33
Axial Bending (Mc)	-128	128	128	-128
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-122	73	202	-116
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14

Shear Stress Summary	10	10	19	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	136	75	204	140
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{102.565(13.179 - 3.0)}$$

$$= 195.629 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	205.130 mm
Nozzle Outside Diameter	(d) =	129.000 mm
Vessel Thickness used	(T) =	10.179 mm
Nozzle Thickness used	(t) =	28.000 mm
T / t	=	0.364
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.823$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.110
M0/P	=	0.034
N0/P	=	0.077
MrD/Mc	=	0.268
NrDL/Mc	=	0.096
M0d/Mc	=	0.081



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Nozzle Calcs.: N4-Shell outlet

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NØDL/Mc = 0.085
MrD/Ml = 0.092
NrDL/Ml = 0.058
MØD/Ml = 0.026
NØDL/Ml = 0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	21	24	21	24
Outplane Stress Summary	18	21	64	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	2	16	119	-78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	22	27	119	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet

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Normal Pressure Stress	10	10	10	10
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Outplane Stress Summary	-23	40	85	-36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	24	21	24

Inplane Stress Summary	-91	127	234	-161
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	91	127	235	162
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	11	9	11	9	11	9	11
Circ. Pl (SUS)	-3	-3	7	7	-3	-3	7	7
Circ. Q (SUS)	-17	17	37	-37	-22	22	42	-42
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-2	-2	6	6	1	1	3	3
Long. Q (SUS)	-15	15	35	-35	-22	22	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	9.0	11.0	9.0	11.0	9.0	11.0	9.0	11.0
Pm+Pl (SUS)	6.2	8.2	16.2	18.1	6.6	8.3	16.1	18.1
Pm+Pl+Q (Total)	18.1	28.2	54.7	27.0	20.5	32.8	61.0	37.6

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N4-Shell outlet

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	18.12	177.00	Passed
Pm+Pl+Q (TOTAL)	61.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	-3	-3	13	13	-5	-5	13	13
Circ. Q (SUS)	0	0	28	-28	-108	108	198	-198
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	9	9	-3	-3	13	13
Long. Q (SUS)	-8	8	98	-98	-32	32	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+Pl (SUS)	18.4	21.3	34.3	37.2	16.4	19.3	34.4	37.3
Pm+Pl+Q (Total)	22.0	26.3	117.2	88.2	92.1	127.1	232.3	161.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+Pl (SUS)	37.28	177.00	Passed
Pm+Pl+Q (TOTAL)	232.33	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	14	11	14	11	14	11	14
Circ. Pl (SUS)	-3	-3	7	7	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-30	-30	36	36	-30	-30	36	36
Long. Q (SUS)	-70	70	130	-130	-98	98	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet

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Shear P1 (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	11.0	14.0	11.0	14.0	11.0	14.0	11.0	14.0
Pm+P1 (SUS)	34.0	36.9	41.7	41.8	37.9	40.8	41.6	41.6
Pm+P1+Q (Total)	109.1	52.8	171.7	111.8	136.5	74.7	200.7	138.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+P1 (SUS)	41.77	177.00	Passed
Pm+P1+Q (TOTAL)	200.74	353.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

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Input, Nozzle Desc: N2-Tube Outlet

From: 80

Pressure for Reinforcement Calculations	P	2401.813	KPa.
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	270	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	188.95	mm
Design Length of Section	L	367.3375	mm
Shell Finished (Minimum) Thickness	t	13.1794	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6586.15	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		89.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.20	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	129.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		20.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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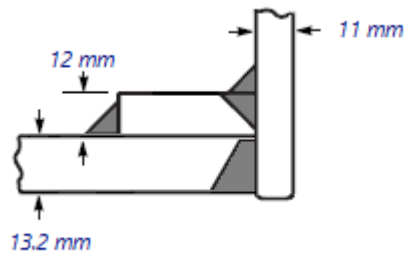
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Nozzle Calcs.: N2-Tube Outlet

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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	89.000 mm.
Actual Thickness Used in Calculation	11.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2402 \cdot 97.5 / (118 \cdot 1 - 0.6 \cdot 2402)$
 $= 2.0087 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2402 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 2402)$
 $= 0.8985 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4125 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.0000 mm
Parallel to Vessel Wall, opening length	d	73.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.4486 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2-Tube Outlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8985 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0087 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.0087 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(5.01, 4.5)]$
 $= 5.0087 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9, 5.01)$
 $= 5.0087 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK



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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	49.2,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	245.8 N./mm ²	Passed
Occasional :	4.9,	Allowable :	156.9 N./mm ²	Passed
Shear :	30.0,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 11, tr = 0.9, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2401.81/10200.00 = 0.235$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11594 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.1366 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.3161 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	188.951	mm
Shell Thickness	13.1794	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	89.000	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	129.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2401.81	KPa.
Radial Load	P	-734.25	Kgf
Circumferential Shear	Vc	530.29	Kgf
Longitudinal Shear	Vl	530.29	Kgf



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Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	130.48	Kg-m.
Torsional Moment	Mt	150.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.337) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (8.091) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{108.565(13.179 - 3.0)}$$

$$= 198.703 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	217.130 mm
Nozzle Outside Diameter	(d) =	89.000 mm
Vessel Thickness used	(T) =	22.179 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	2.772
d / t	=	11.125

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.282$$

Nr/P	=	0.144
Mr/P	=	0.118
M0/P	=	0.115
N0/P	=	0.184
MrD/Mc	=	0.184
NrDL/Mc	=	0.178
M0d/Mc	=	0.187
N0DL/Mc	=	0.055
MrD/Ml	=	0.147
NrDL/Ml	=	0.146
M0D/Ml	=	0.158
N0DL/Ml	=	0.173

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	9	11	9	11
Outplane Stress Summary	-10	27	54	-18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2



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Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-12	18	47	-25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane	Shear	(Vc)	1	1	-1	-1
Outplane	Shear	(Vl)	0	0	0	0
Outplane	Shear	(Mt)	5	5	5	5
Shear Stress Summary			7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	18	30	56	26

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4
Outplane Stress Summary	-16	28	51	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		2	2	2	2
Inplane Bending (P)		10	-10	10	-10
Inplane Membrane (Mc)		-5	-5	5	5
Inplane Bending (Mc)		-32	32	32	-32
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		9	11	9	11
Inplane Stress Summary		-15	30	59	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5



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Shear Stress Summary	3	3	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	33	63	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	9	12	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-33	-33	33	33
Axial Bending (ML)	-100	100	100	-100
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-93	45	174	-87

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	19	19	10	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	A1	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	54	174	111

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0



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Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14

Hoop Stress Summary	12	15	16	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-33	-33	33	33
Axial Bending (Mc)	-128	128	128	-128
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-122	73	202	-116
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14

Shear Stress Summary	10	10	19	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	136	75	204	140
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 89.0 + 2 \cdot 1.65 \cdot \sqrt{102.565(13.179 - 3.0)}$$

$$= 195.629 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	205.130 mm
Nozzle Outside Diameter	(d) =	129.000 mm
Vessel Thickness used	(T) =	10.179 mm
Nozzle Thickness used	(t) =	28.000 mm
T / t	=	0.364
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.823$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.110
M0/P	=	0.034
N0/P	=	0.077
MrD/Mc	=	0.268
NrDL/Mc	=	0.096
M0d/Mc	=	0.081



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NØDL/Mc = 0.085
MrD/Ml = 0.092
NrDL/Ml = 0.058
MØD/Ml = 0.026
NØDL/Ml = 0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	21	24	21	24
Outplane Stress Summary	18	21	64	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	2	16	119	-78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	22	27	119	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Normal Pressure Stress	10	10	10	10
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Outplane Stress Summary	-23	40	85	-36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	24	21	24

Inplane Stress Summary	-91	127	234	-161
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	8	8
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	91	127	235	162
------------------------	----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	11	9	11	9	11	9	11
Circ. Pl (SUS)	-3	-3	7	7	-3	-3	7	7
Circ. Q (SUS)	-17	17	37	-37	-22	22	42	-42
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-2	-2	6	6	1	1	3	3
Long. Q (SUS)	-15	15	35	-35	-22	22	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	9.0	11.0	9.0	11.0	9.0	11.0	9.0	11.0
Pm+Pl (SUS)	6.2	8.2	16.2	18.1	6.6	8.3	16.1	18.1
Pm+Pl+Q (Total)	18.1	28.2	54.7	27.0	20.5	32.8	61.0	37.6

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N2-Tube Outlet

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	18.12	177.00	Passed
Pm+Pl+Q (TOTAL)	61.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	-3	-3	13	13	-5	-5	13	13
Circ. Q (SUS)	0	0	28	-28	-108	108	198	-198
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	9	9	-3	-3	13	13
Long. Q (SUS)	-8	8	98	-98	-32	32	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+Pl (SUS)	18.4	21.3	34.3	37.2	16.4	19.3	34.4	37.3
Pm+Pl+Q (Total)	22.0	26.3	117.2	88.2	92.1	127.1	232.3	161.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+Pl (SUS)	37.28	177.00	Passed
Pm+Pl+Q (TOTAL)	232.33	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	14	11	14	11	14	11	14
Circ. Pl (SUS)	-3	-3	7	7	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-30	-30	36	36	-30	-30	36	36
Long. Q (SUS)	-70	70	130	-130	-98	98	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Shear P1 (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	11.0	14.0	11.0	14.0	11.0	14.0	11.0	14.0
Pm+P1 (SUS)	34.0	36.9	41.7	41.8	37.9	40.8	41.6	41.6
Pm+P1+Q (Total)	109.1	52.8	171.7	111.8	136.5	74.7	200.7	138.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+P1 (SUS)	41.77	177.00	Passed
Pm+P1+Q (TOTAL)	200.74	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	2400.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2400.00 KPa.
Shell Minimum Operating Pressure	Psox,min	15.00 KPa.
Shell Thickness	ts	15.0622 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	188.951 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	270.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	2400.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	2400.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	15.0622 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	188.951 mm
Channel Design Temperature	TEMPC	260.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	81
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	12.7000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5860.00 mm
Design Temperature of the Tubes		260.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.30 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	146.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	15.8750 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	80.203 mm



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ASME TS Calc:

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	163.7 cm ²
Perimeter of Tube Layout	Cp	476.2 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	260.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	134.60 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	70.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	360.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	243.00 mm
Step 1 Depth on the Tubeside	ht1	9.00 mm
Step 2 Diameter on the Tubeside	dt2	192.00 mm
Step 2 Depth on the Tubeside	ht2	7.00 mm
Is the Tubesheet Flat on Shellside?		Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4		243.00 mm

Note: Tubesheet diameter is now: 243.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	560.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-103.4190 KPa.
Channel Side Min. Design Pressure	Ptd,min	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	360.000 mm
Flange Inside Diameter	B	188.951 mm
Flange Face Outside Diameter	Fod	257.951 mm
Flange Face Inside Diameter	Fid	188.951 mm
Gasket Outside Diameter	Go	246.951 mm
Gasket Inside Diameter	Gi	220.951 mm
Small end Hub thk.	g0	13.1794 mm
Large end Hub thk.	g1	20.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	3.1750 mm



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ASME TS Calc:

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Full face Gasket Flange Option

Program Selects

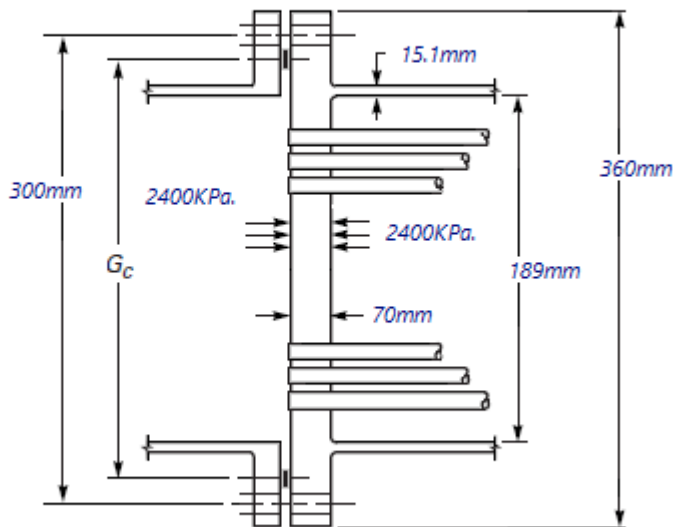
Bolting Information:

Diameter of Bolt Circle	C	300.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	UNC Thread Series	
Number of Bolts	n	12
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 54.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	270.0 °C	0.18723E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%



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Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	260.0 °C	0.18822E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(2503 \cdot (12.7/2 - 0)) / (92.3 \cdot 1 + 0.4 \cdot 2503)$
= 0.1704 + 0.0000 = 0.1704 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 260.00 °C
Elastic Modulus for Material 186156960.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	12.70	5860.00	6.02	50.0000	0.0303634	90.33

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 24994$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3994	12.70	5860.00	31.80	50.0000	0.0010880	59.71

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 59.7) / (3 \cdot 31.8) = 2504$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3994 mm
Allowable Internal Pressure at Corroded thickness	35368.44 KPa.
Required Internal Design Pressure	2503.42 KPa.
Allowable External Pressure at Corroded thickness	24994.28 KPa.
Required External Design Pressure	2503.42 KPa.
Required Thickness due to Shell Side pressure	0.3994 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm



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Gasket Reaction Diameter, $G = Go - 2.0 \cdot b$ 234.107 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= l_{tx} / h$ (modified for corrosion if present)

$= 0/70$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

$= \max(12.7 - 2 \cdot 2.11 (188224240/188224240) \cdot$

$(92.3/135) \cdot (0); 12.7 - 2 \cdot 2.11)$

$= 12.7000$ mm

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 \cdot ro + dt$

$= 2 \cdot 80.2 + 12.7$

$= 173.106$ mm

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$

$= (15.9 - 12.7) / 15.9$

$= 0.200$

Equivalent outer tube limit radius $[ao]$:

$= Do / 2$

$= 173/2$

$= 86.5530$ mm

Effective tube pitch $[p^*]$:

$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

$= 15.9 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 173 \cdot 15.9) / (\pi \cdot 173^2)) }$

$= 15.8750$ mm

Effective ligament efficiency for bending $[\mu^*]$:

$= (p^* - d^*) / p^*$

$= (15.9 - 12.7) / 15.9$

$= 0.2000$

Ratio $[ps]$:

$= as / ao$

$= 94.5/86.6$

$= 1.0915$

Ratio $[pc]$:

$= ac / ao$

$= 117/86.6$

$= 1.3524$

Compute parameter $[xt]$:

$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

$= 1 - 81 ((12.7 - 2 \cdot 2.11) / (2 \cdot 86.6))^2$

$= 0.8056$

Determine parameter $[xs]$:

$= 1 - Nt(dt / (2 \cdot ao))^2$

$= 1 - 81 (12.7 / (2 \cdot 86.6))^2$

$= 0.5640$



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ASME TS Calc:

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Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts)E_s/L$$

$$= \pi \cdot 15.1 (189 + 15.1)187232032/5860$$

$$= 31453.9180 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt)E_t/L$$

$$= \pi \cdot 2.11 (12.7 - 2.11)188224240/5860$$

$$= 229.9337 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 31454 / (81 \cdot 230)$$

$$= 1.68883$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_t)$$

$$= 1 / (1 + 31454/0)$$

$$= 1.0000 \text{ (= 1 if No Exp. Jt.)}$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((189 + 15.1) 15.1)$$

$$= 0.0328 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.033 \cdot 187232032 \cdot 15.1^3 / (6(1 - 0.3^2))$$

$$= 3842690048.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(189 + 15.1) \cdot 3.84E+09/70^3 \cdot (1 + 70 \cdot 0.033 + (70 \cdot 0.033)^2/2)$$

$$= 81324392.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$$

$$= 189^2 / (4 \cdot 187232032 \cdot 15.1) \cdot (1 - 189 \cdot 0.3 / (2(189 + 15.1)))$$

$$= 0.27251E-02 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 4.409449$; $\mu^* = 0.200000$; $E^* = 29240470 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (4.409) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})N_t \cdot E_t \cdot tt(d_t - tt)a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2)81 \cdot 188224240 \cdot 2.11$$

$$(12.7 - 2.11)86.6^2) / (29240470 \cdot 5860 \cdot 70^3))^{1/4}$$

$$= 0.9525$$

Values from Table 4.18.3

$Z_d = 2.482893$; $Z_v = 0.170039$; $Z_m = 0.681610$



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ASME TS Calc:

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$$Z_a = 0.350844E+00 ; Z_w = 0.170039$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 243/173 \\ &= 1.4038 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 29240470 \cdot (81324392 + 0 + 188224240 \cdot \ln(1.4)) \\ &= 2.6879 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*) F \\ &= (1 + 0.46) 2.69 \\ &= 3.9205 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.09 - 1 - 3.92 \cdot 0.17) / (1 + 3.92 \cdot 0.68) \\ &= -0.156608 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (2.48 + -0.16 \cdot 0.17) / 2 \cdot 0.95^4 \\ &= 1.0108 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.17 + -0.16 \cdot 0.68) / 2 \cdot 0.95^4 \\ &= 0.0260 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.17 + (1.09 - 1) 0.68) 0.95^4 / (1 + 3.92 \cdot 0.68) \\ &= 0.0521 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (234 - 300) / 173 = -0.38 \end{aligned}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \psi_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.09 \cdot 3.84E+09 \cdot 0.033 \cdot 0.002725107 (1 + 70 \cdot 0.033) \\ &= 1235.3409 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s \\ &= 86.6^2 (1.09^2 - 1) \cdot (1.09 - 1) / 4 - 1235 \\ &= -1202.5232 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \psi_c \cdot k_c [\delta_c (C_m \theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta_c / 2)] \\ &= 1.35 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 70) - 0.000 (0/70 + 0 \cdot 0/2)] \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$



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Parameter [$\omega \cdot c$]:

$$= ao^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c$$

$$= 86.6^2 [(1.35^2 + 1) \cdot (1.35 - 1) / 4 - (1.09 - 1) / 2] - 0$$

$$= 1524.1832 \text{ mm}^2$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= Ps \{ xs + 2(1 - xs) vt + [2 / Kst \cdot (Ds / Do)^2] vs - [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2 J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \}$$

$$= 2400 \{ 0.56 + 2(1 - 0.56) 0.3 + [2 / 1.69 \cdot (189 / 173)^2] 0.3 - [(1.09^2 - 1) / (1 \cdot 1.69)] - [(1 - 1) / (2 \cdot 1 \cdot 1.69)] [(0^2 - (189)^2) / 173^2] \}$$

$$= 2725.2944 \text{ KPa.}$$

Pressure [$P't$]:

$$= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt$$

$$= [0.81 + 2(1 - 0.81) 0.3 + 1 / (1 \cdot 1.69)] 2400$$

$$= 3634.4932 \text{ KPa.}$$

Pressure [P_Y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 81 \cdot 230 \cdot 0 / (3.14 \cdot 86.6^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$$

$$= -0.38 \cdot 0.052 \cdot 74816 / (2 \cdot \pi \cdot 86.6^2)$$

$$= 309.0623 \text{ KPa.}$$

Calculate the pressure [$Prim$]:

$$= - (U / ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= - (0.052 / 86.6^2) \cdot (-1203 \cdot 2400 - 1524 \cdot 2400)$$

$$= 45.5040 \text{ KPa.}$$

Calculate the pressure [$P\omega$]:

$$= (U / ao^2) \cdot (\omega s \cdot P's - \omega c \cdot P \cdot c)$$

$$= (0.052 / 86.6^2) \cdot (1235 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho s - 1) Qz2)) \cdot (P's - P't + P_Y + P\omega + P_w + Prim)$$

$$= 0.1000E+01 \cdot 1.69 / (1 + 1 \cdot 1.69 (1.01 + (1.09 - 1) 0.026)) \cdot (2725 - 3634 + 0 + 0 + 309 + 45.5)$$

$$= -345.5055 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega s \cdot P's - \omega c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-1203 \cdot 2400 - 1524 \cdot 2400) - (1235 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 74816 \cdot -0.38 / (2 \cdot \pi)] / (1 + 3.92 \cdot 0.68)$$

$$= -1416.0040 \text{ Kg}$$

Calculate factor [$Q3$]:



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$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\
 &= -0.16 + 2 \cdot -1416 / (-346 \cdot 86.6^2) \\
 &= 10.57287
 \end{aligned}$$

Fm Value from Table 4.18.3 = 5.424430

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 5.42 / 0.2) \cdot (2 \cdot 86.6 / (70 - 0))^2 \cdot -346 \\
 &= -85.9654 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 135 = 202 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 1.36 / 0.2) \cdot (2 \cdot 86.6 / (27 - 0))^2 \cdot 482 \\
 &= 201.8940 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 27 + 0 + 0 \\
 &= 27.0014 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.2 \cdot 70 / 86.6) \cdot 108 \\
 &= 34.8302 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ ulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned}
 \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\
 \text{Tubesheet Shear Stress} &= 16.7185 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(27; 3.81) \\
 &= 27.0014 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = 0.24, Ftmax = 1.76$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\
 &= ((2400 \cdot 0.56 - 2400 \cdot 0.81) - (-346) \cdot 0.24) / \\
 &\quad (0.81 - 0.56) \\
 &= -2.0619 \text{ N./mm}^2
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\
 &= ((2400 \cdot 0.56 - 2400 \cdot 0.81) - (-346) \cdot 1.76) / \\
 &\quad (0.81 - 0.56) \\
 &= 0.1191 \text{ N./mm}^2
 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.06 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= Sot = 92.2986 \text{ N./mm}^2$$



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Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(2.48 + 10.6 \cdot 0.17) 0.95^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (\frac{d_t^2}{4} + (\frac{d_t}{2} - t_t)^2) }) / 4$$

$$= (\sqrt{ (\frac{12.7^2}{4} + (\frac{12.7}{2} - 2.11)^2) }) / 4$$

$$= 3.8177 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 188224240 / 146)}$$

$$= 159.3101$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 560 / 3.82$$

$$= 146.6842$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (146 / 2) \cdot (1 - 147 / (2 \cdot 159))$$

$$= 39.501 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.06 \cdot 0.7$$

$$= 14.76 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (12.7 - 2.11) 2.11$$

$$= 0.702 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 92.3$$

$$= 660.685 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (12.7 + 0.67 \cdot 2.11) 92.3$$

$$= 660.6851 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.3 / 92.3$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (12.7 + 0.67 \cdot 2.11) 92.3$$

$$= 484.2863 \text{ Kg}$$



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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 12.7)^2 + 1.07 \cdot 2.11 \cdot (12.7 - 2.11) \cdot 1 \cdot 1} - 0.75 \cdot 12.7 \right)$$

$$= 2.364 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2660 mm

Minimum Required Groove Weld Leg [agr]: 1.2660 mm

Tube-Tubesheet Jt allowable, 661 is >= tube strength 661 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= Ss \cdot Esw = 118 \cdot 1 = 118 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σs,m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 86.6^2 / ((189 + 15.1) \cdot 15.1) \cdot [-346 + (1.09^2 - 1)(2400 - 2400)] + 94.5^2 \cdot 2400 / ((189 + 15.1) \cdot 15.1)$$

$$= 6.1292 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 15.1 mm, the minimum Shell length adjacent to the tubesheet should be 96.026 mm

Shell Membrane Stress due to Joint Interaction [σs,m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 86.6^2 / ((189 + 15.1) \cdot 15.1) \cdot [-346 + (1.09^2 - 1)(2400 - 2400)] + 94.5^2 \cdot 2400 / ((189 + 15.1) \cdot 15.1)$$

$$= 6.1292 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σs,b]:

$$= 6 \cdot ks / ts^2 \cdot \{ \beta s [\delta s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) \cdot (ao/h)^3 \cdot (1 + h \cdot \beta s / 2) \cdot [Pe (Zv + Zm \cdot Q1) + 2 \cdot ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 3.84E+09 / 15.1^2 \cdot \{ 0.033 [0.002725107 \cdot 2400 + 94.5^2 \cdot 0 / (2.82E+09)] + 6(1 - 0.46^2) / (29240470) \cdot (86.6/70)^3 \cdot (1 + 70 \cdot 0.033/2) \cdot [-346 (0.17 + 0.68 \cdot -0.16) + 2/86.6^2 \cdot 0.68 \cdot -1416] \}$$

$$= -148.6209 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot Ss$$

$$\therefore |6.13| + |-149| \leq 177 \text{ N./mm}^2$$

$$\therefore 155, \text{ must be } \leq 177 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	86.0 <=	201.9 N./mm ²	Pass
Tubesheet Shear Stress	0.8 <=	107.7 N./mm ²	Pass
Maximum Tube Stress	2.1 <=	92.3 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.1 <=	-39.5 N./mm ²	Pass
Maximum Force on any one Tube	14.8 <=	660.7 Kgf	Pass
Axial Membrane Stress in Shell	6.1 <=	118.0 N./mm ²	Pass
Shell Stress (jt. inter.)	154.8 <=	177.0 N./mm ²	Pass



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Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	27.001	70.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.266	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.266	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA	Tbsht	Extnsn	---- Tubesheet		Stresses		Case Type	Pass/Fail	
	Bend			Allwd	Shear	Allwd				
D1uc	9.286		37.706	92	202	5	108	Ps+Pt-Th	D1	Ok
D2uc	12.291		...	11	202	4	108	Ps+Pt-Th	D2	Ok
D3uc	27.001		...	86	202	1	108	Ps+Pt-Th	D3	Ok
D4uc	3.810		...	1	202	...	108	Ps+Pt-Th	D4	Ok
O1uc	9.956		...	92	404	5	108	Ps+Pt+Th	O1	Ok
O2uc	17.766		...	86	404	5	108	Ps+Pt+Th	O2	Ok
O3uc	9.519		...	86	404	1	108	Ps+Pt+Th	O3	Ok
O4uc	11.011		...	75	404	...	108	Ps+Pt+Th	O4	Ok
D1c	46.033		37.706	121	202	5	108	Ps+Pt-Th-c	D1	Ok
D2c	17.660		...	12	202	5	108	Ps+Pt-Th-c	D2	Ok
D3c	44.664		...	111	202	1	108	Ps+Pt-Th-c	D3	Ok
D4c	9.810		...	1	202	...	108	Ps+Pt-Th-c	D4	Ok
O1c	19.099		...	119	404	5	108	Ps+Pt+Th-c	O1	Ok
O2c	28.959		...	110	404	5	108	Ps+Pt+Th-c	O2	Ok
O3c	25.349		...	111	404	1	108	Ps+Pt+Th-c	O3	Ok
O4c	25.054		...	99	404	1	108	Ps+Pt+Th-c	O4	Ok
Max:	46.0334		37.706	mm	0.600		0.051	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+/-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses	----	:	----	Shell Band Stress	----	:	Pass
Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	: Fail



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D1uc	1	117	...	...	...	...	...	...	Ok
D2uc	5	117	...	...	...	...	...	...	Ok
D3uc	6	117	...	...	...	...	...	...	Ok
D4uc	...	117	...	-108	...	...	...	...	Ok
O1uc	1	389	...	...	...	...	...	...	Ok
O2uc	6	389	...	...	...	...	...	...	Ok
O3uc	6	389	...	...	...	...	...	...	Ok
O4uc	...	389	...	...	...	...	...	...	Ok
D1c	1	117	...	...	...	...	...	...	Ok
D2c	7	117	...	...	...	...	...	...	Ok
D3c	8	117	...	...	...	...	...	...	Ok
D4c	...	117	...	-105	...	...	...	...	Ok
O1c	1	389	...	...	...	...	...	...	Ok
O2c	8	389	...	...	...	...	...	...	Ok
O3c	8	389	...	...	...	...	...	...	Ok
O4c	1	389	...	...	...	...	...	...	Ok

Max RATIO 0.069 0.003 ...

Tube, Shell and Channel Stress Summary:

		---- Tube Stresses		Tube Loads		Shell Stress		Channel	Stress Pass		
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	2	92	-1	-39	12	660	170	176	...	...	Ok
D2uc	2	92	-2	-39	11	660	26	176	...	...	Ok
D3uc	2	92	-2	-39	15	660	155	176	...	...	Ok
D4uc	...	92	...	...	...	660	...	176	...	...	Ok
O1uc	2	184	-1	-39	12	1321	169	389	...	...	Ok
O2uc	3	184	-3	-39	23	1321	133	389	...	...	Ok
O3uc	2	184	-2	-39	15	1321	155	389	...	...	Ok
O4uc	2	184	-2	-39	12	1321	147	389	...	...	Ok
D1c	2	92	-1	-39	13	660	219	389	...	...	Ok
D2c	1	92	-1	-39	9	660	36	176	...	...	Ok
D3c	3	92	-3	-39	19	660	197	389	...	...	Ok
D4c	...	92	...	-39	...	660	...	176	...	...	Ok
O1c	2	184	-1	-39	12	1321	218	389	...	...	Ok
O2c	4	184	-4	-39	26	1321	168	389	...	...	Ok
O3c	3	184	-3	-39	19	1321	197	389	...	...	Ok
O4c	2	184	-2	-39	18	1321	189	389	...	...	Ok

Max RATIO 0.028 0.091 0.028 0.958 ...

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :			
	46.033	70.000	Ok
Tubesheet thickness flanged extension :			
	37.706	54.000	Ok
Tube thickness :			
	0.399	2.110	Ok
Tube-tubesheet fillet weld leg :			
	1.266	2.110	Ok
Tube-tubesheet groove weld leg :			
	1.266	2.110	Ok

Min Shell length of thk, (15.062) adj. to tubesheet: 96.026 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.068 mm
 Due to Pressure + Thermal Shell Compresses by : -0.068 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force	Tubeside	0 shellside	Shellside	0 tubeside
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ASME TS Calc:

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Condition	MAWP	Stress Rat.	MAWP	Stress Rat.
Tubesheet Bending Stress	9740.23	1.000	42122.71	1.000
Tubesheet Shear Stress	43194.61	1.000	58257.62	1.000
Tube Tensile Stress	141007.59	1.000	148292.67	1.000
Tube Compressive Stress	No Calc	No Calc	62592.63	1.000
Tube-Tubesheet Joint load	141007.59	1.000	148289.88	1.000
Shell Stress (Axial, Junction)	9740.23	1.000	24994.27	0.930
Tube Pressure Stress	35368.43	1.000	24994.27	1.000
Tubesheet Extension Stress	10277.60	...	No Calc	No Calc
Minimum MAWP	9740.23		24994.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	13055.24	1.000	44887.56	1.000
Tubesheet Shear Stress	48985.20	1.000	63069.34	1.000
Tube Tensile Stress	146968.98	1.000	148085.11	1.000
Tube Compressive Stress	No Calc	No Calc	73287.18	1.000
Tube-Tubesheet Joint load	146968.98	1.000	148085.14	1.000
Shell Stress (Axial, Junction)	13055.24	1.000	26520.31	0.570
Tube Pressure Stress	35406.67	1.000	26520.31	1.000
Tubesheet Extension Stress	10277.60	...	No Calc	No Calc
Minimum MAPnc	10277.60		26520.31	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0068	0.0068	...	...	...	...
2	-0.0675	-0.0676	...	...	...	...
3	-0.0635	-0.0636	...	...	...	...
4	0.0027	0.0027	...	...	...	...
5	0.0039	0.0039	...	...	...	...
6	-0.0674	-0.0674	...	...	...	...
7	-0.0635	-0.0636	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, shell thickness))

= max(70/4, min(70, 15.1))

= 17.500 mm

Thickness Ratio = 0.23, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:

= tubesheet thickness/4

= 70/4



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ASME TS Calc:

Case: 1 9:39a Sep 9, 2026

= 17.500 mm

Thickness Ratio = 0.23, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(2400/10278, 2400/26520)

= 0.234

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Problems/Failures Summary:

Step: 32

9:39am

Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: